



Computer Networks

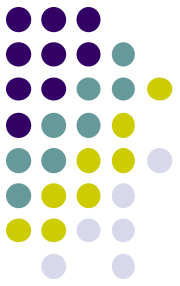
CS3611

Introduction-Part 2

Haiming Jin

The slides are adapted from those provided by Prof. Shizhen Zhao and Romit Roy Choudhury.

Chapter 1: Introduction

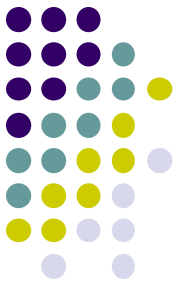


Our goal:

- get “feel” and terminology
- paint a broad picture
- see the forest through the trees
- approach:
 - use Internet as example

Topics:

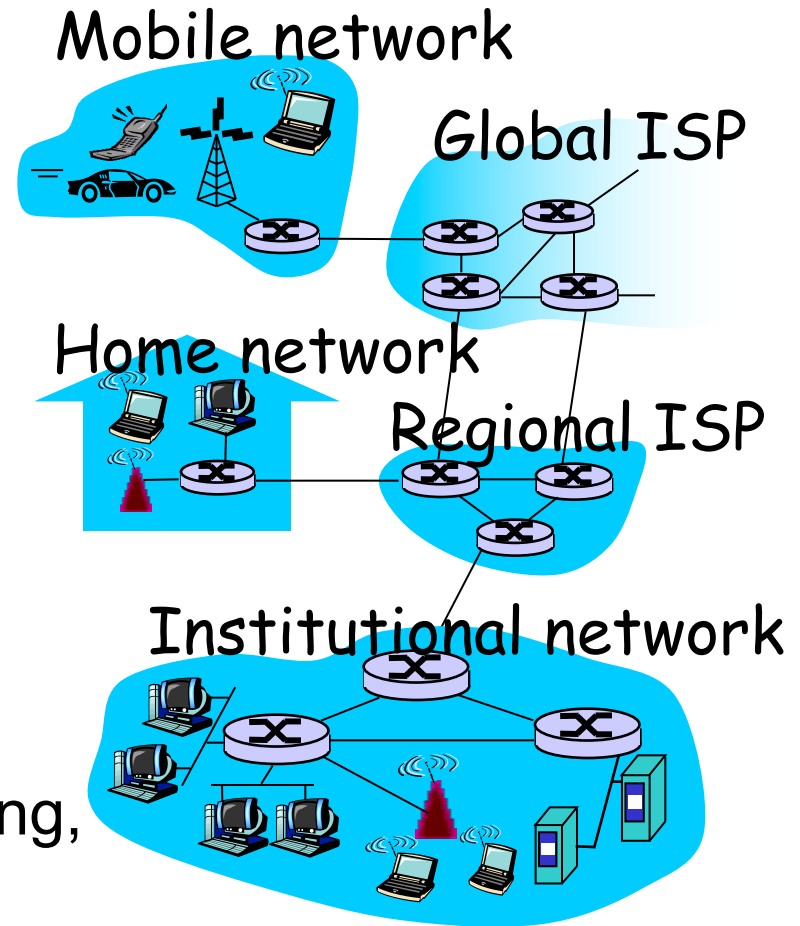
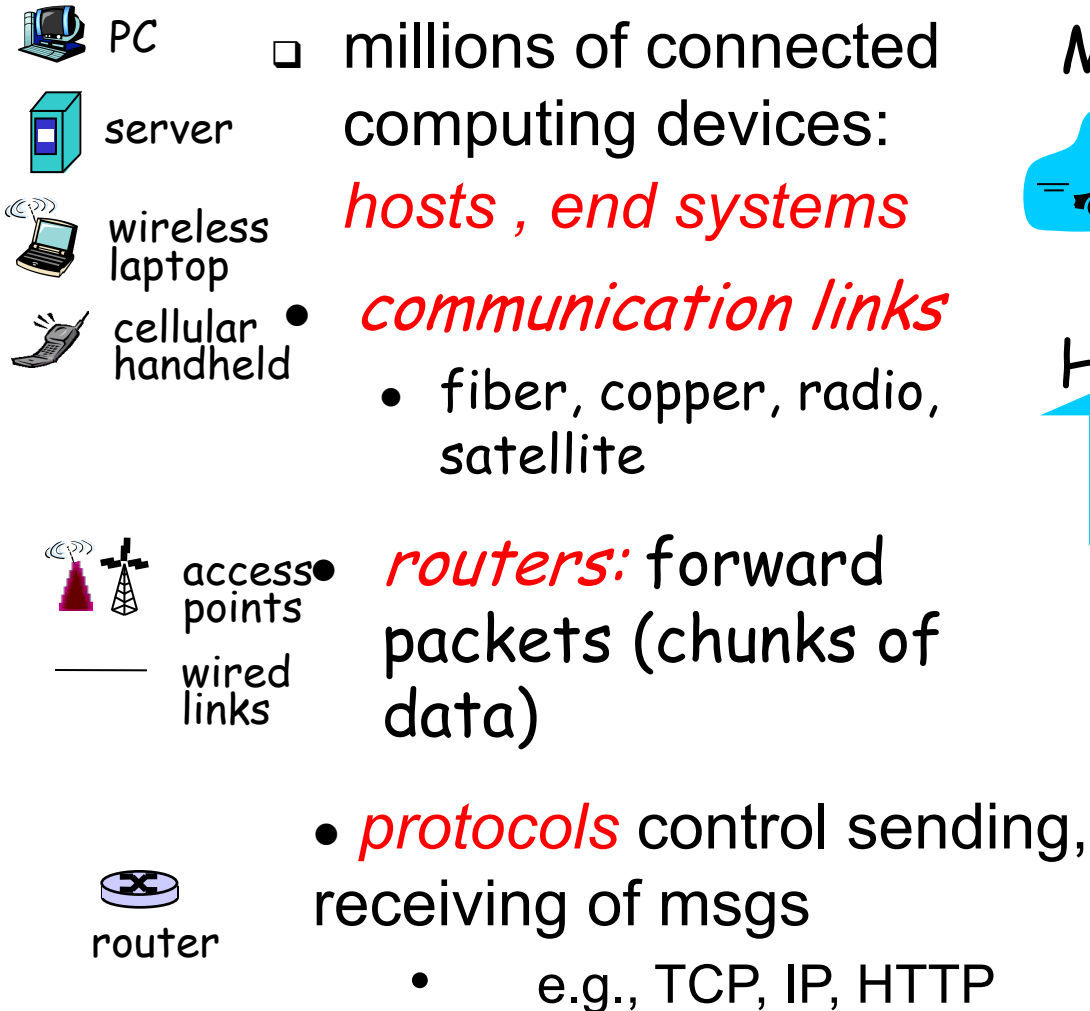
- What’s Computer Network?
- Protocol layers, service models
- What’s the Internet?
- Network edge
- Access net and physical media
- Network core
- Internet structure and ISPs
- Delay, loss, and throughput in packet-switched networks
- History of Internet



Chapter 1: roadmap

- What's Computer Network?
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What's the Internet: "nuts and bolts" view



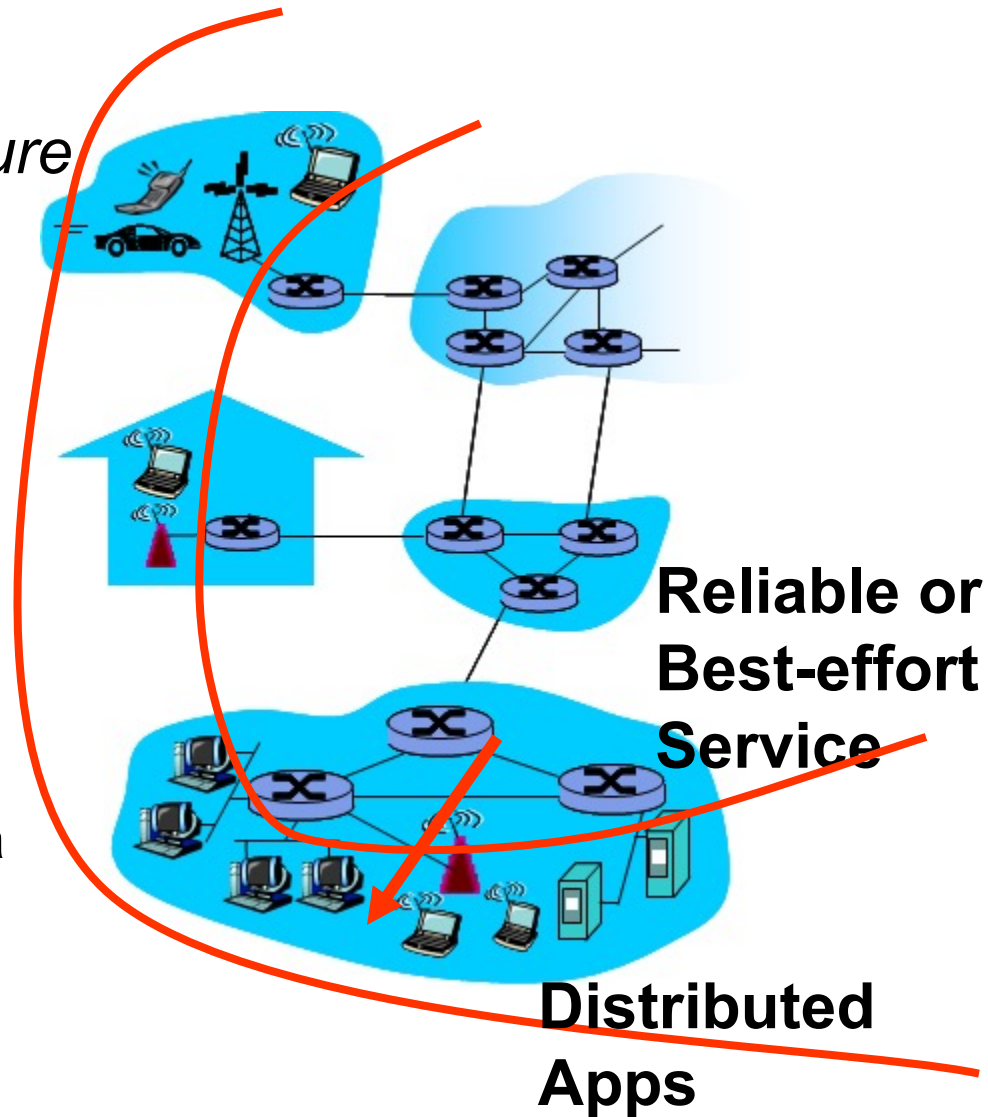
What's the Internet: a service view

□ communication *infrastructure* enables **distributed applications**:

- Web, VoIP, email, games, e-commerce, file sharing

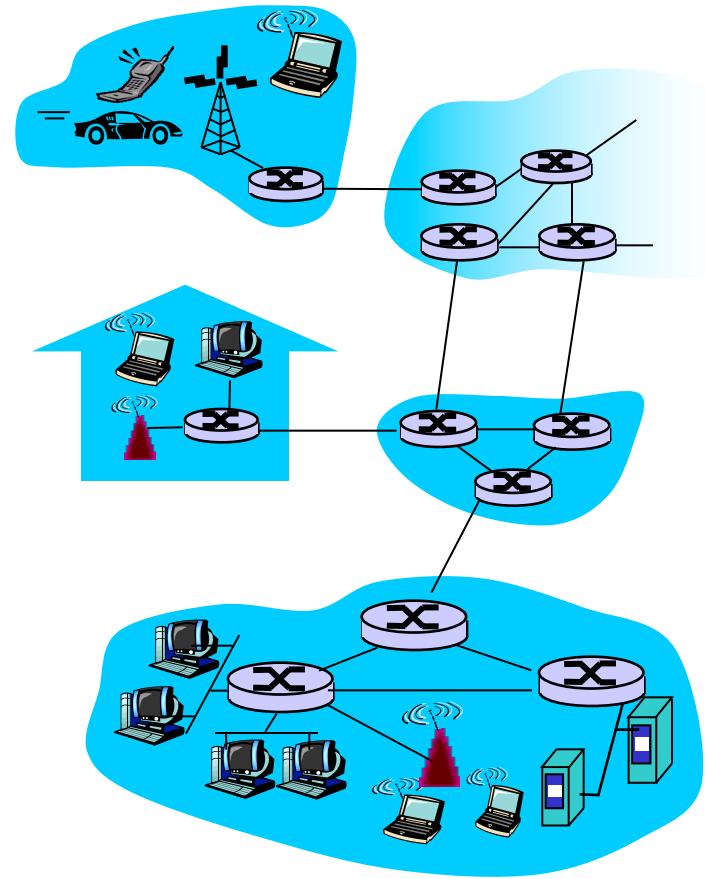
□ **communication services** provided to apps:

- reliable data delivery from source to destination
- “best effort” (unreliable) data delivery



Internet Structure

- **network edge:**
applications and hosts
- **access networks,**
physical media: wired,
wireless communication
links
- **network core:**
 - ❖ interconnected
routers
 - ❖ network of networks



Chapter 1: roadmap

- ❑ What's Computer Network?
- ❑ Protocol layers, service models
- ❑ What's the Internet?
- ❑ **Network edge**
- ❑ Access net and physical media
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The network edge

□ end systems (hosts):

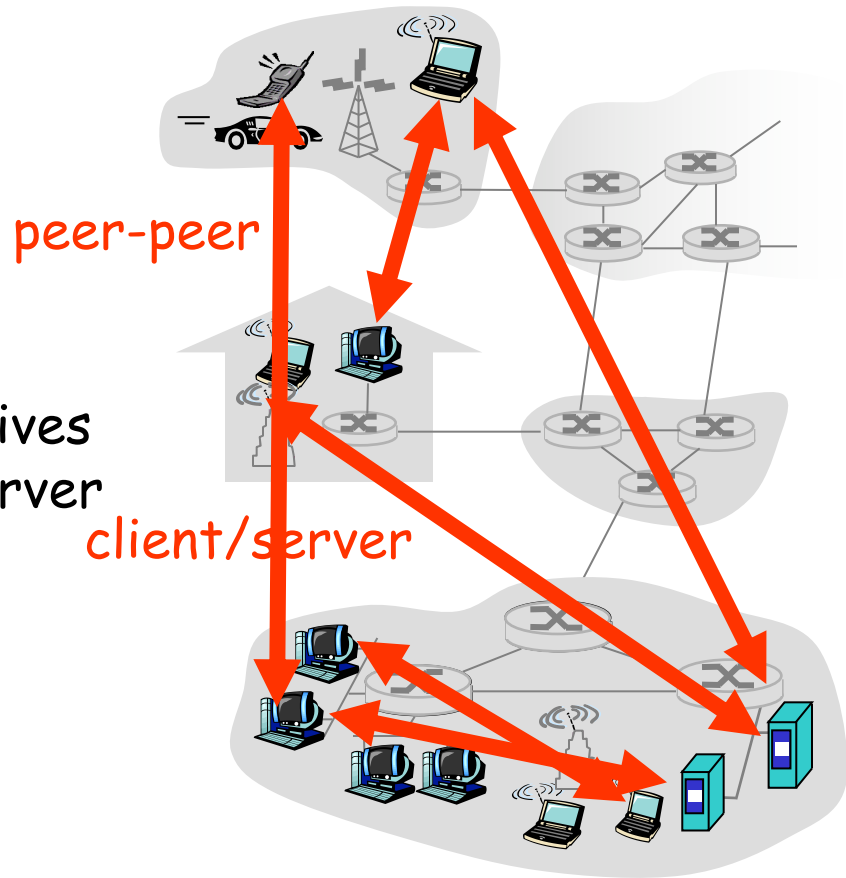
- run application programs
- e.g. Web, email
- at “edge of network”

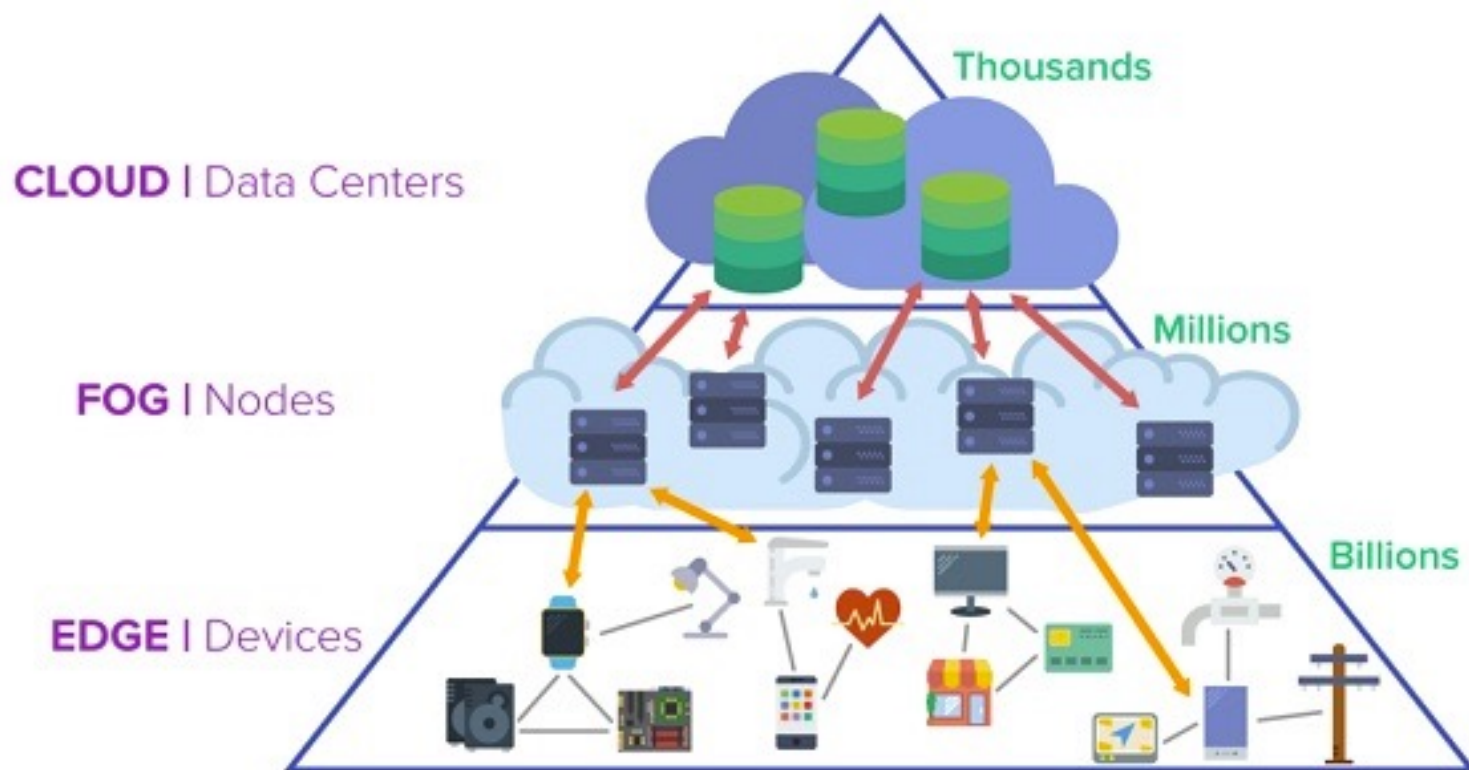
● client/server model

- ❖ client host requests, receives service from always-on server
- ❖ e.g. Web browser/server; email client/server

● peer-peer model:

- ❖ minimal (or no) use of dedicated servers
- ❖ e.g. Skype, BitTorrent





Network edge: connection-oriented service

Goal: data transfer between end systems

- ❑ **Connection:** prepare for data transfer ahead of time
 - Request / Respond
 - *set up “state”* in two communicating hosts
- ❑ TCP - Transmission Control Protocol
 - Internet's connection-oriented service

TCP service [RFC 793]

- ❑ *reliable, in-order* byte-stream data transfer
 - loss: acknowledgements and retransmissions
- ❑ *flow control:*
 - sender won't overwhelm receiver
- ❑ *congestion control:*
 - senders “slow down sending rate” when network congested

Network edge: connectionless service

Goal: data transfer between end systems

- same as before!

❑ **UDP** - User Datagram Protocol [RFC 768]:

- connectionless
- unreliable data transfer
- no flow control
- no congestion control

App's using TCP:

- ❑ HTTP (Web), FTP (file transfer), Telnet (remote login), SMTP (email)

App's using UDP:

- ❑ streaming media, teleconferencing, DNS, Internet telephony

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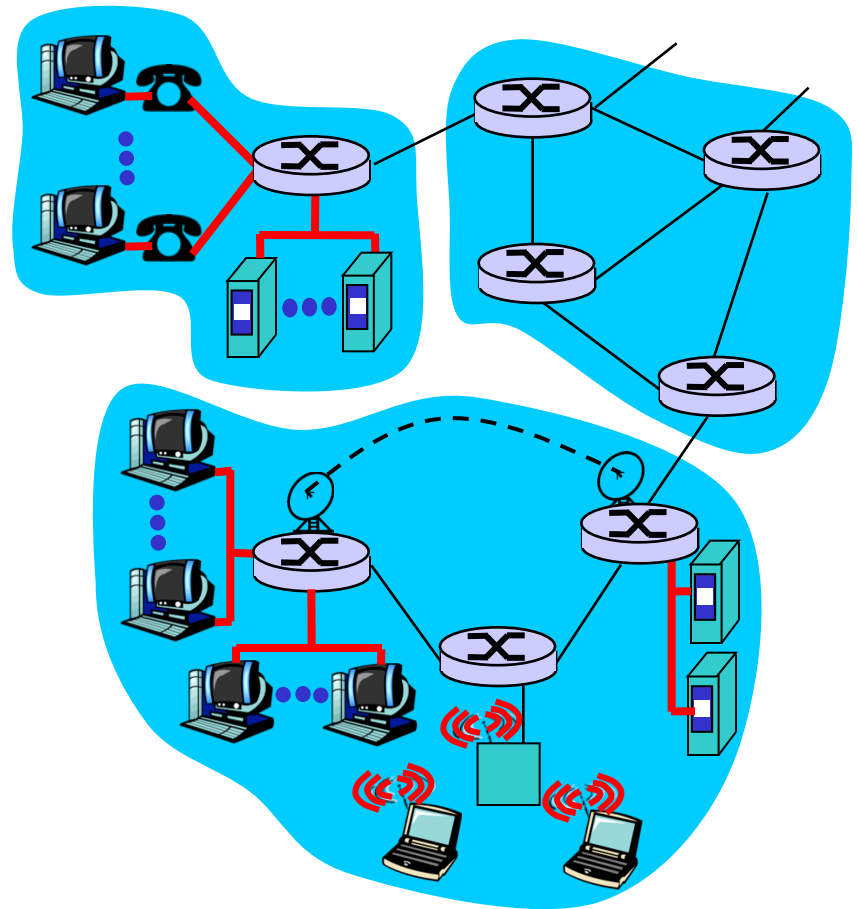
Access networks and physical media

Q: How to connect end systems to edge router?

- ❑ residential access nets
- ❑ institutional access networks (school, company)
- ❑ mobile access networks

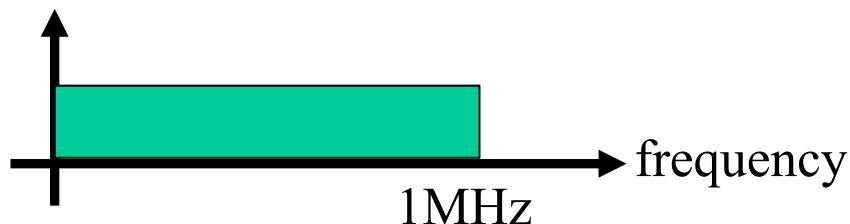
Keep in mind:

- ❑ bandwidth of access network?
- ❑ shared or dedicated?



Bandwidth vs Data Rate

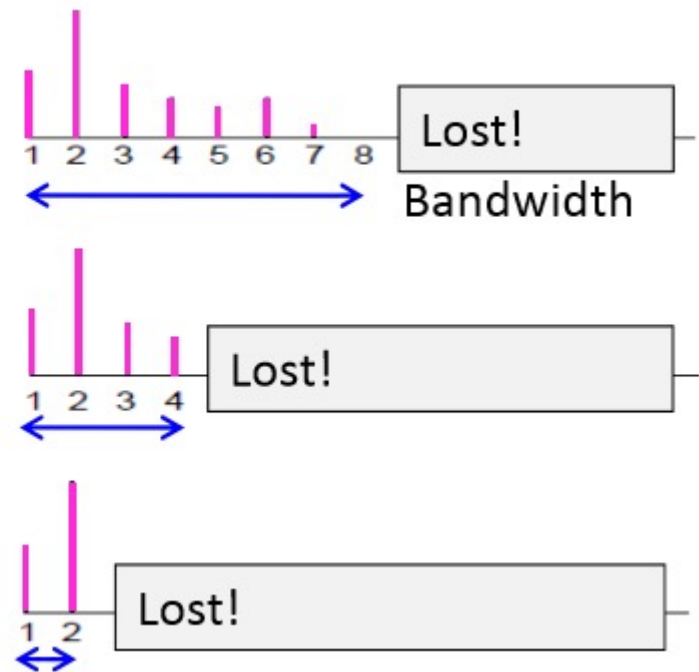
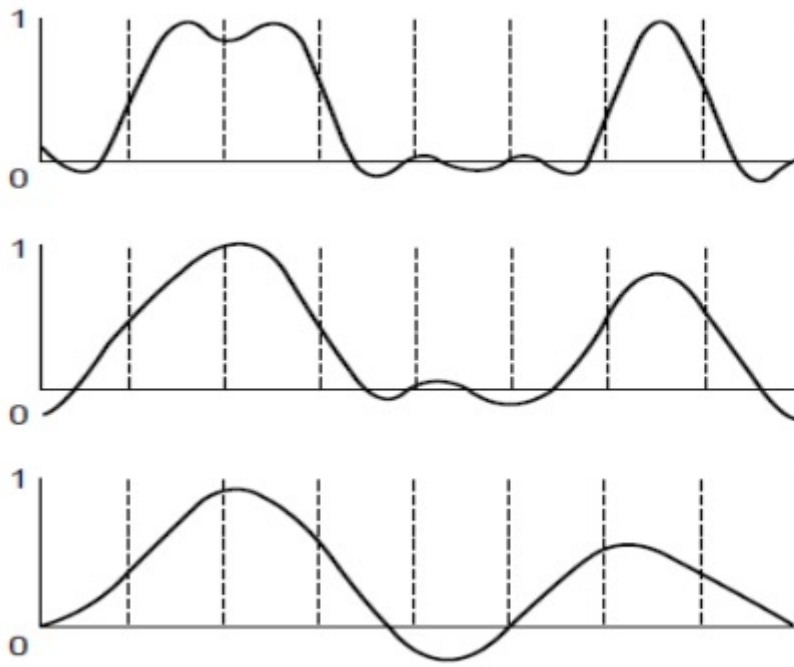
- **Bandwidth (Hz):** the range of frequencies transmitted without being strongly attenuated.
- The bandwidth is a physical property of the transmission medium and usually depends on the construction, thickness, and length of the medium.



Bandwidth=1MHz

Bandwidth vs Data Rate

- ❑ A Wide Band signal will be **distorted** when transmitted thru relatively narrower band channel with the higher harmonics cut off or hold back.



Bandwidth vs Data Rate

□ **Data Rate (aka Bit Rate) (bit/s)**: the rate at which bits can be transmitted.

□ **Shannon's theorem**: the maximum data rate of a noisy channel with signal-to-noise ratio S/N is:

$$R = B \log_2(1 + S/N)$$

The diagram shows the equation $R = B \log_2(1 + S/N)$ with two arrows pointing from variables to their units. An arrow points from R to the text "Data rate (bit/s)", and another arrow points from B to the text "Bandwidth (Hz)".

Data rate (bit/s)

Bandwidth (Hz)

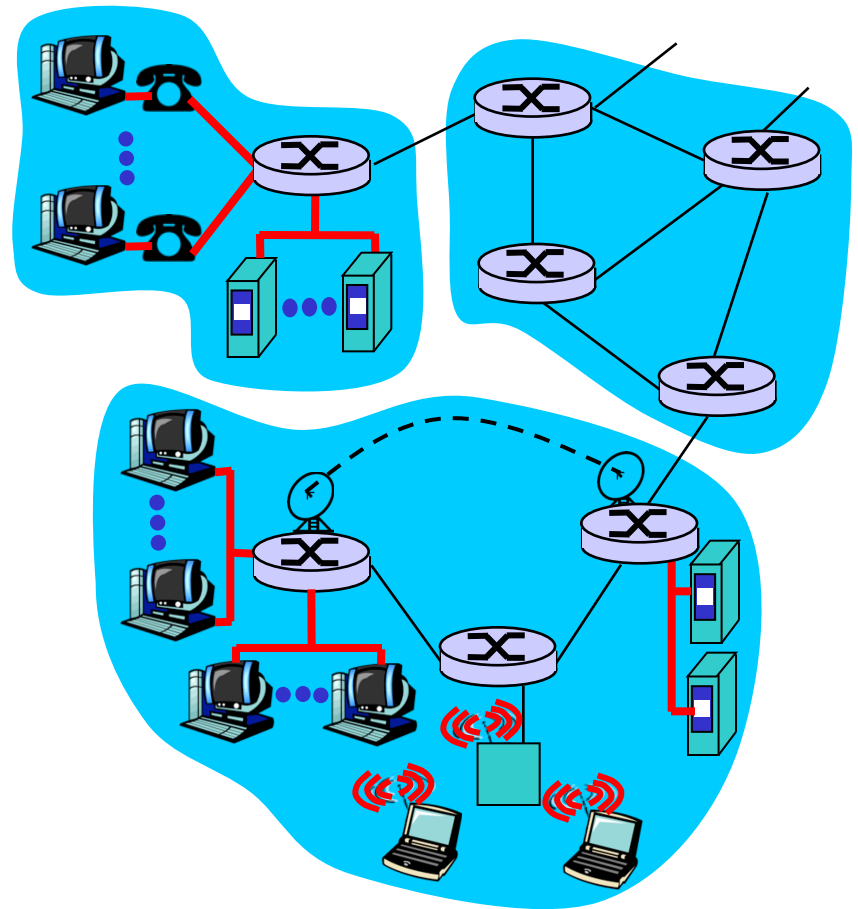
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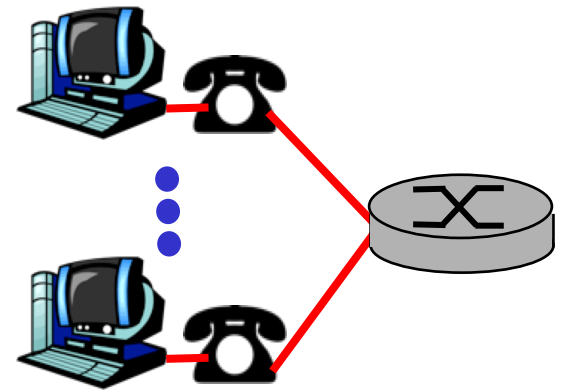
- ❑ bandwidth of access network?
- ❑ shared or dedicated?



Residential access: point to point access

❑ Dialup via modem

- up to 56Kbps direct access to router (often less)
- Can't surf and phone at same time: can't be “always on”



❑ ADSL: asymmetric digital subscriber line

- up to 1 Mbps upstream
- up to 8 Mbps downstream
- FDM: 50 kHz - 1 MHz for downstream
4 kHz - 50 kHz for upstream
0 kHz - 4 kHz for ordinary telephone

Residential access: cable modems

- ❑ HFC: hybrid fiber coax

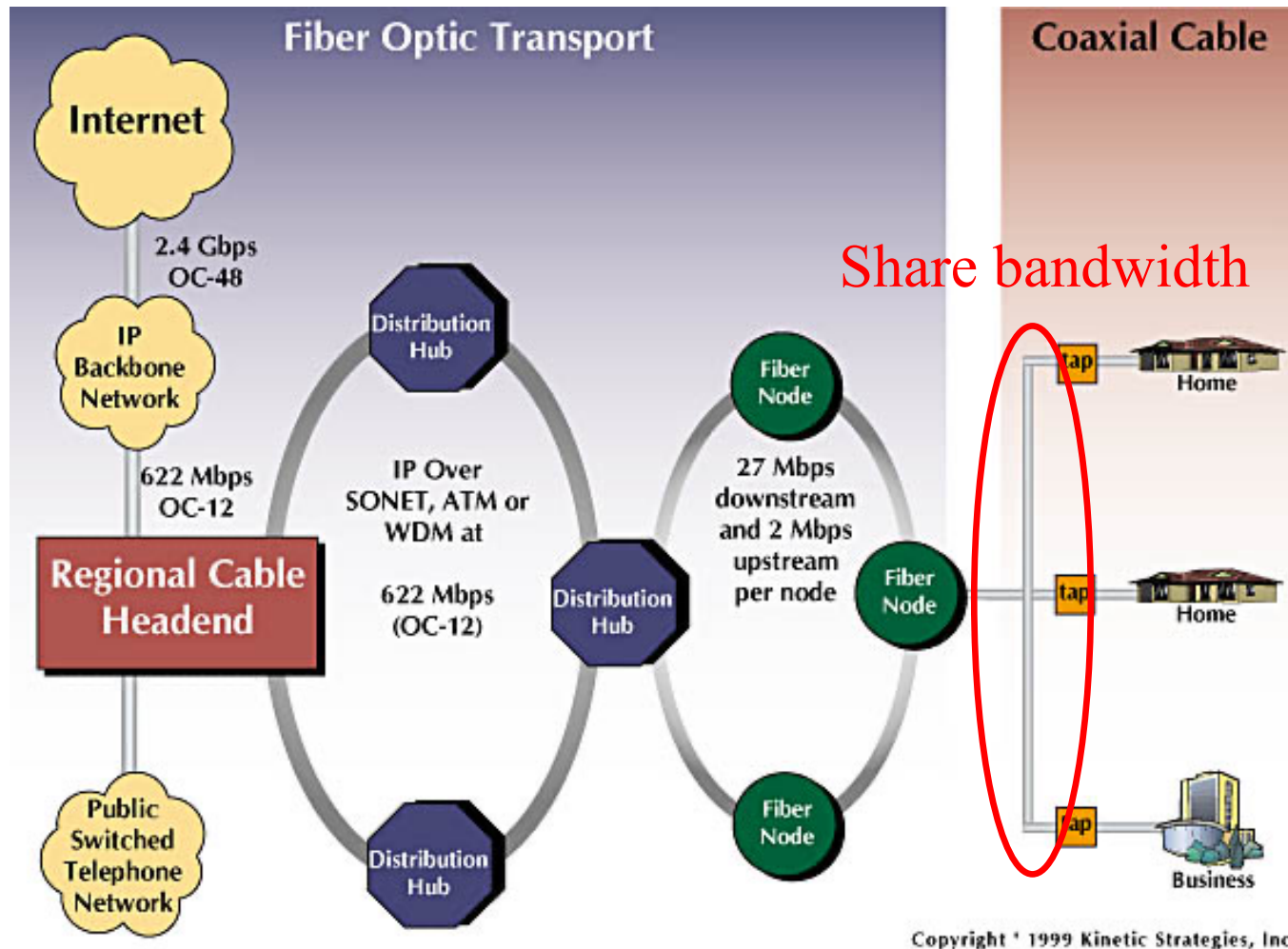
- asymmetric: up to 30Mbps downstream, 2 Mbps upstream

- ❑ network of cable and fiber attaches home to ISP router

- homes share access to router

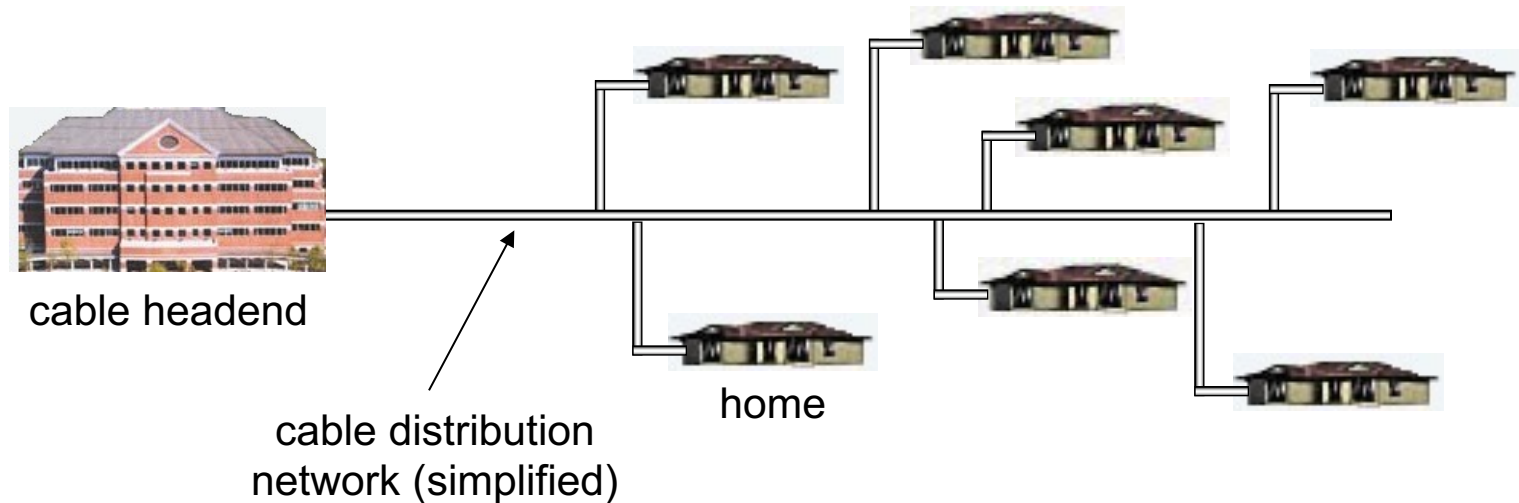
- ❑ deployment: available via cable TV companies

Residential access: cable modems

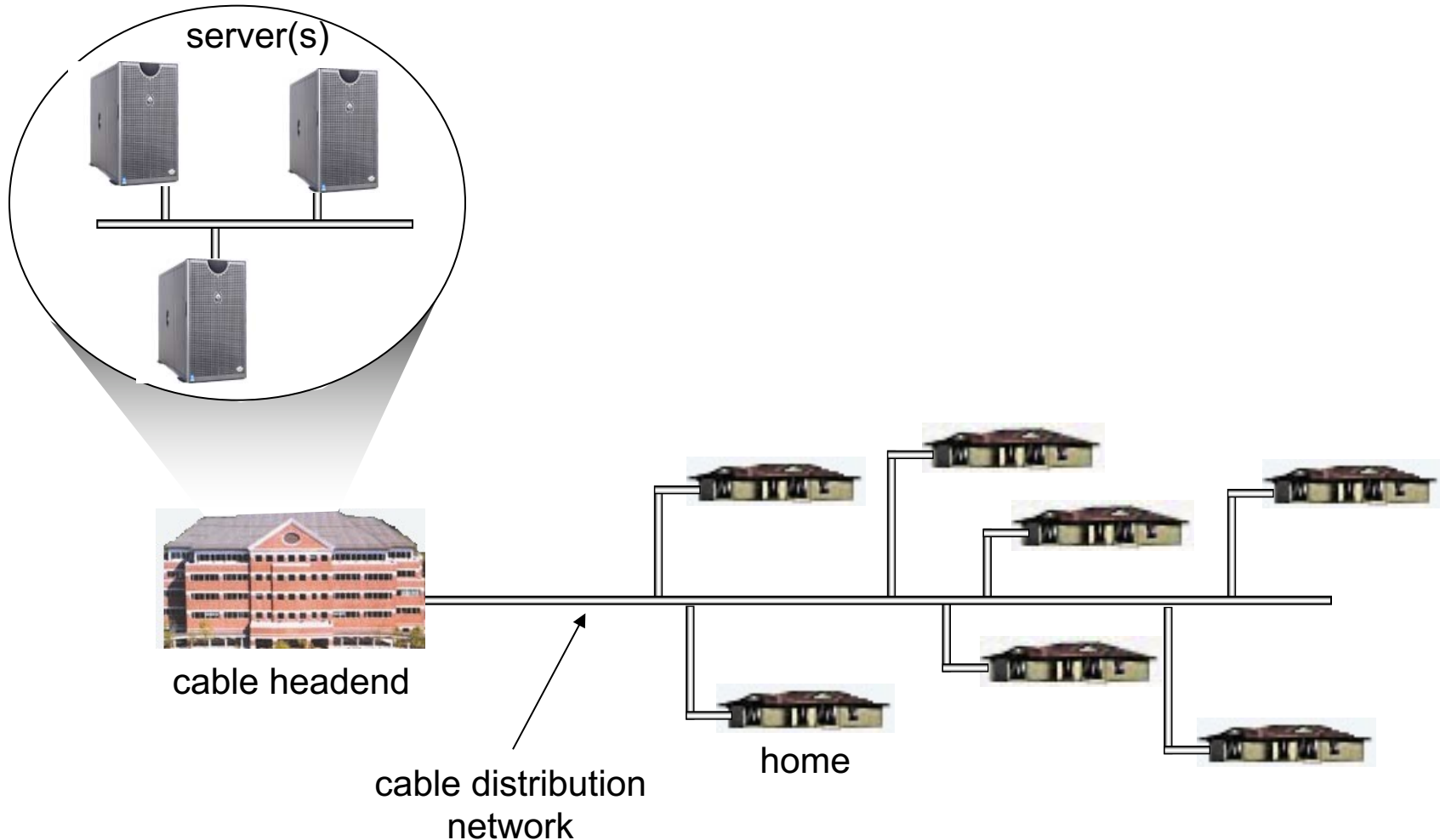


Cable Network Architecture: Overview

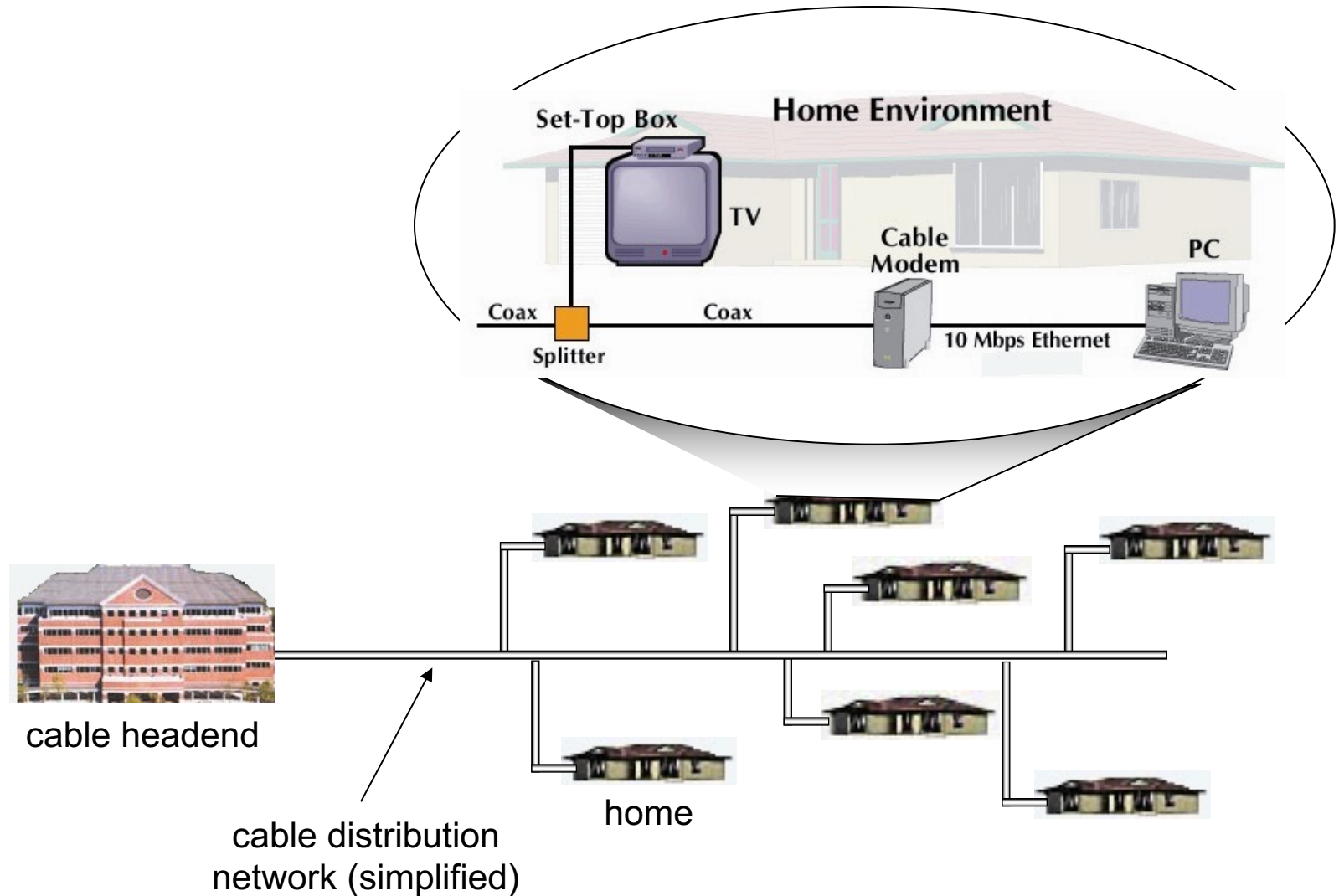
Typically 500 to 5,000 homes



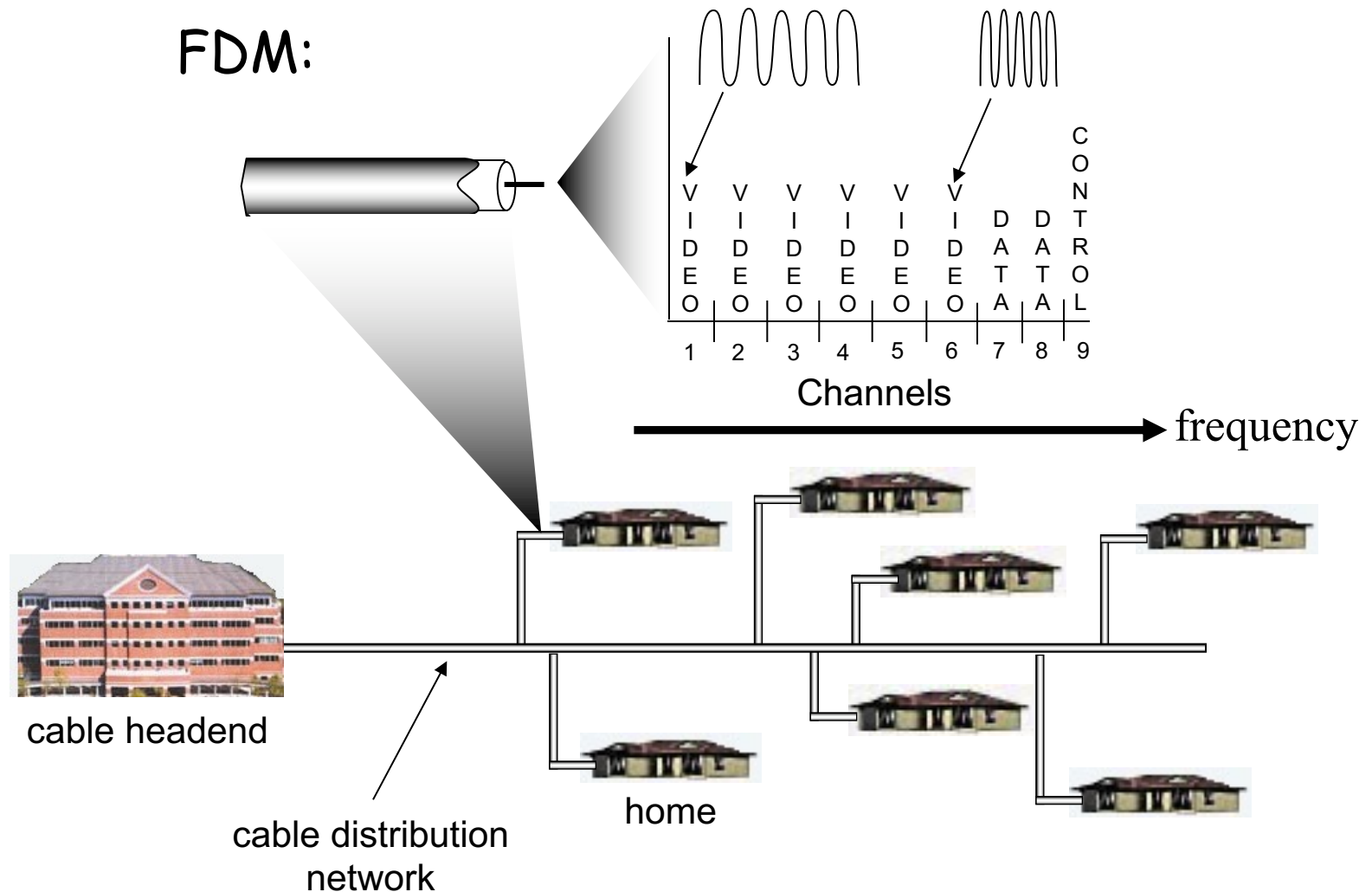
Cable Network Architecture: Overview



Cable Network Architecture: Overview



Cable Network Architecture: Overview



DSL vs Cable Modem

- ❑ DSL is point to point

Thus data rate does not reduce when neighbor uses his/her DSL

- ❑ But, DSL uses twisted-pair, and transmission technology cannot support more than ~10Mbps

- ❑ Cable Modems share the pipe to the cable headend.

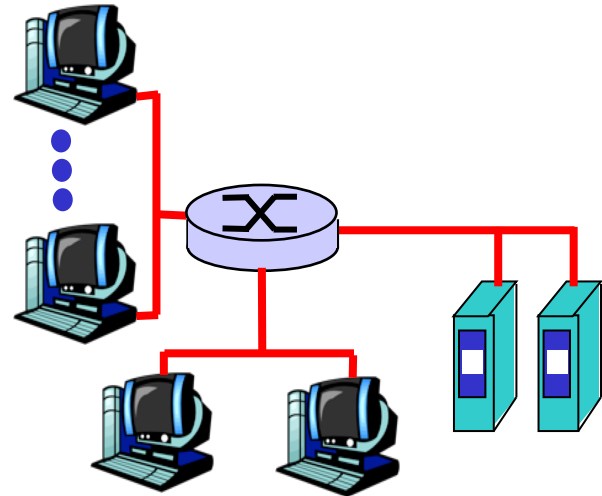
Thus, your data rate can reduce when neighbors are surfing concurrently

- ❑ However, fibre optic lines have significantly higher data rate (fat pipe)

Even if other users, data rate may still be higher

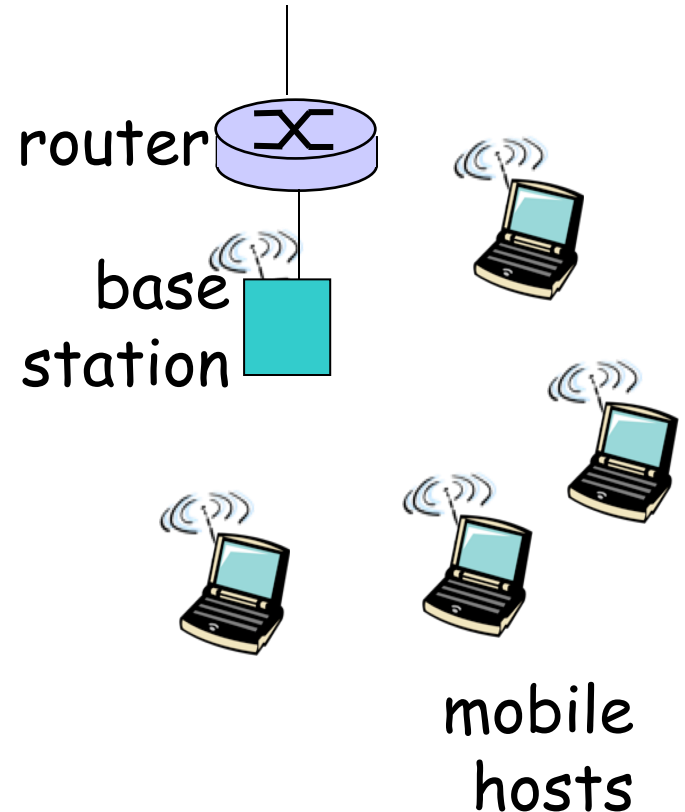
Company access: local area networks

- ❑ company/univ **local area network** (LAN) connects end system to edge router
- ❑ **Ethernet:**
 - shared or dedicated link connects end system and router
 - 10 Mbs, 100Mbps, Gigabit Ethernet



Wireless access networks

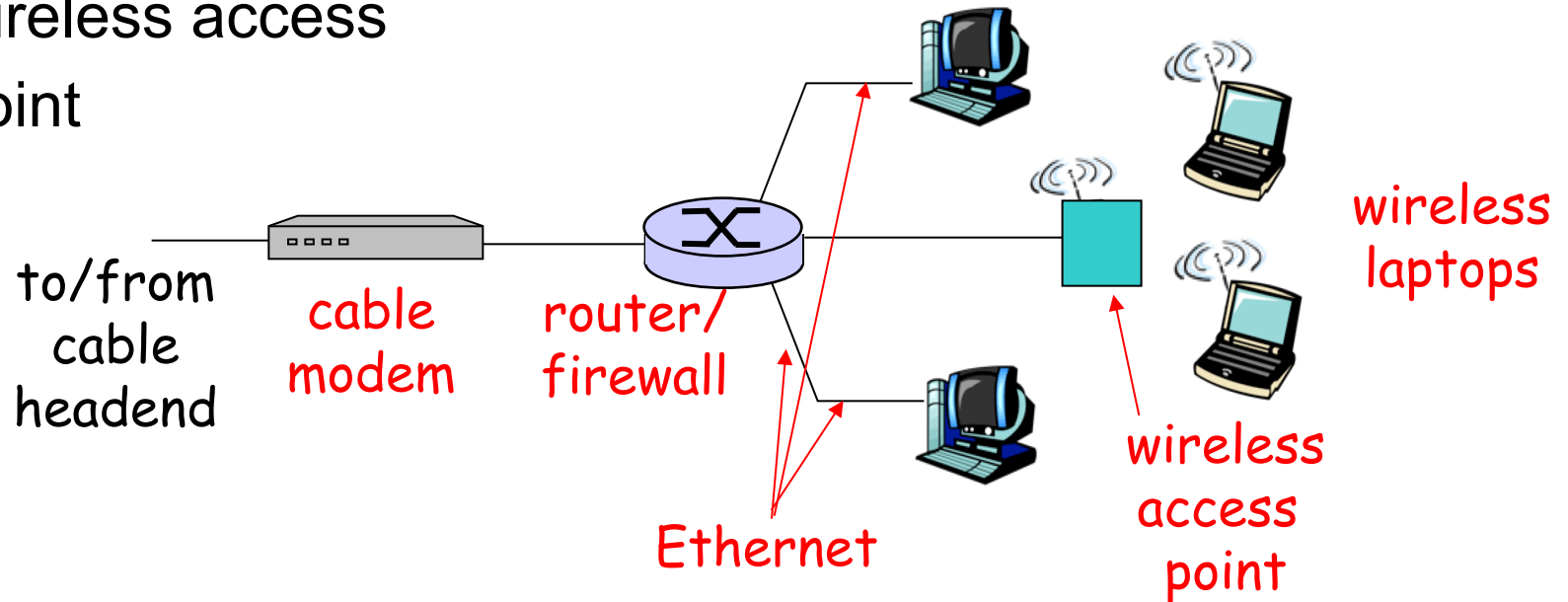
- ❑ shared *wireless* access network connects end system to router
 - via base station aka “access point”
- ❑ **wireless LANs:**
 - 802.11b/g (WiFi): 11 or 54 Mbps
- ❑ **wider-area wireless access**
 - provided by telco operator



Home networks

Typical home network components:

- ❑ ADSL or cable modem
- ❑ router/firewall/NAT
- ❑ Ethernet
- ❑ wireless access point



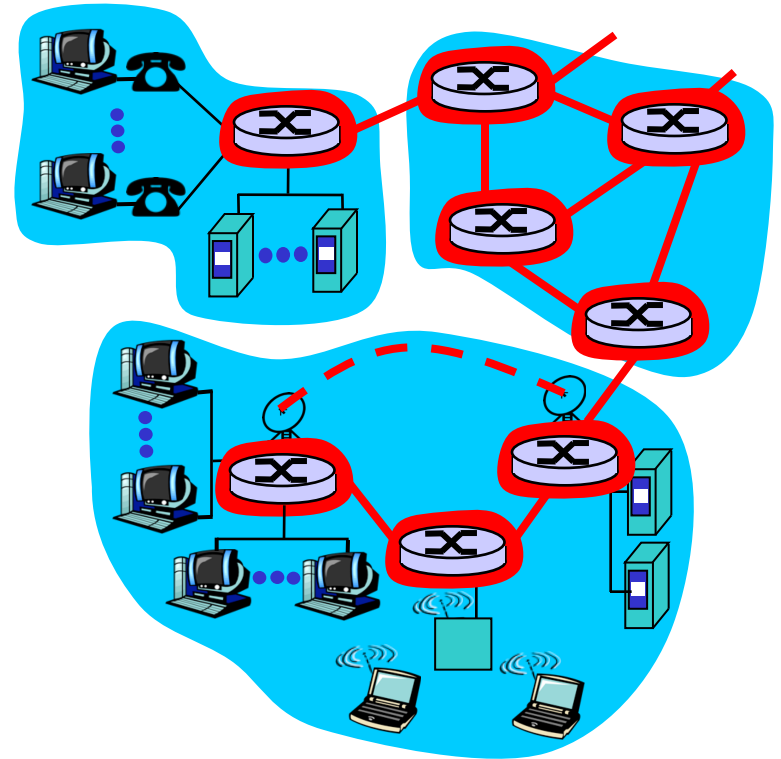
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The Network Core

- ❑ mesh of interconnected routers
- ❑ the fundamental question: how is data transferred through net?

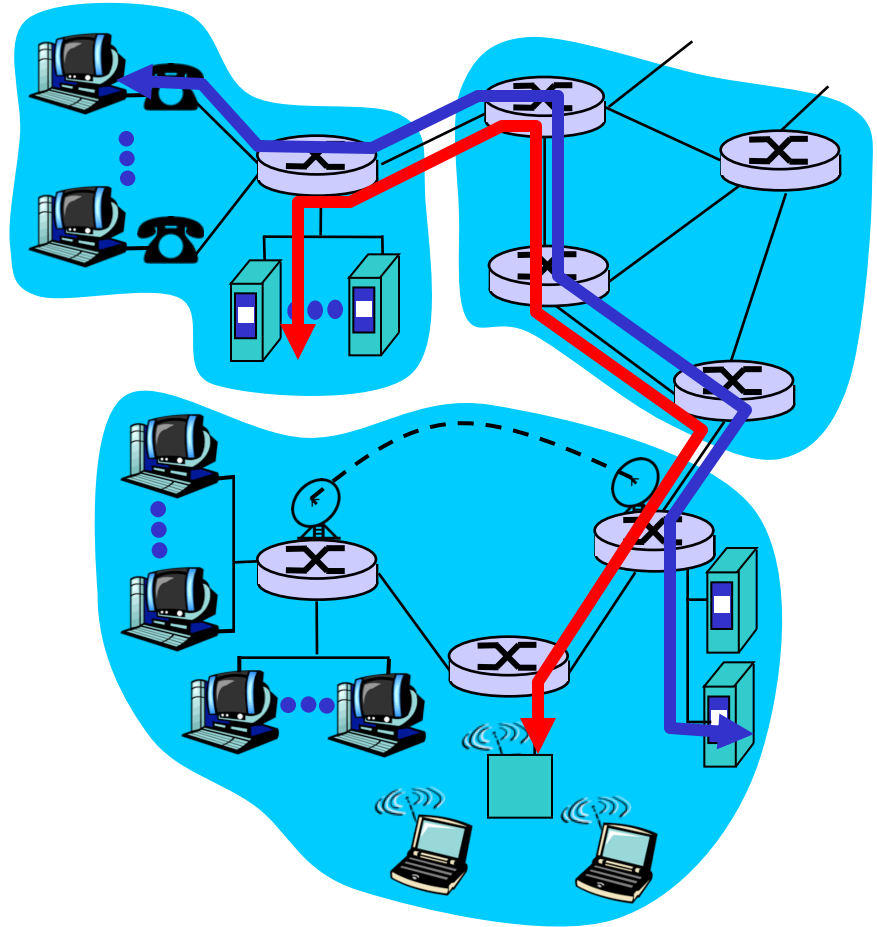
- **circuit switching:** dedicated circuit per call: telephone net
- **packet-switching:** data sent thru net in discrete “chunks”



Network Core: Circuit Switching

End-end resources
reserved for “call”

- ❑ link bandwidth, switch capacity
- ❑ dedicated resources: no sharing
- ❑ circuit-like (guaranteed) performance
- ❑ call setup required



Network Core: Circuit Switching

network resources (e.g., bandwidth) **divided into “pieces”**

- pieces allocated to calls

- resource piece **idle** if not used by owning call (*no sharing*)

- dividing link bandwidth into “pieces”

- frequency division
- time division

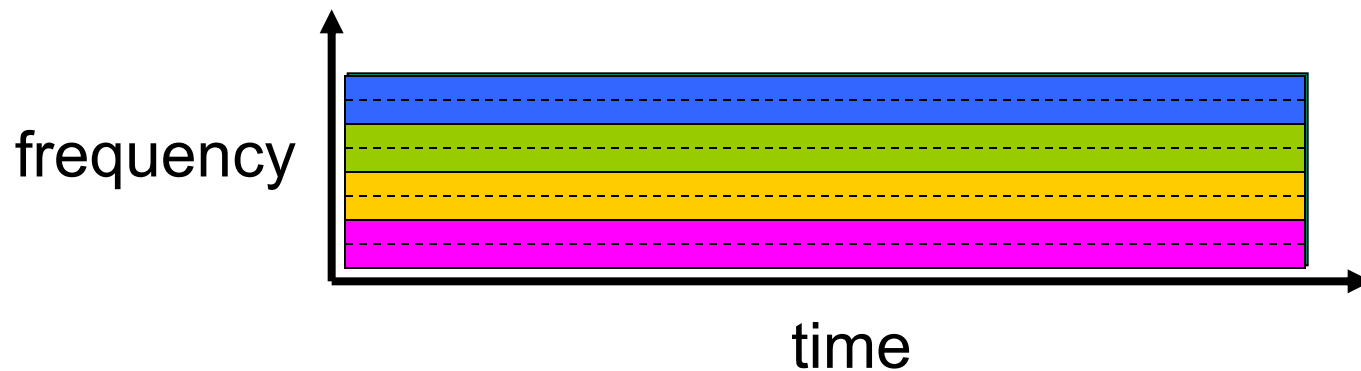
Circuit Switching: FDM and TDM

Example:

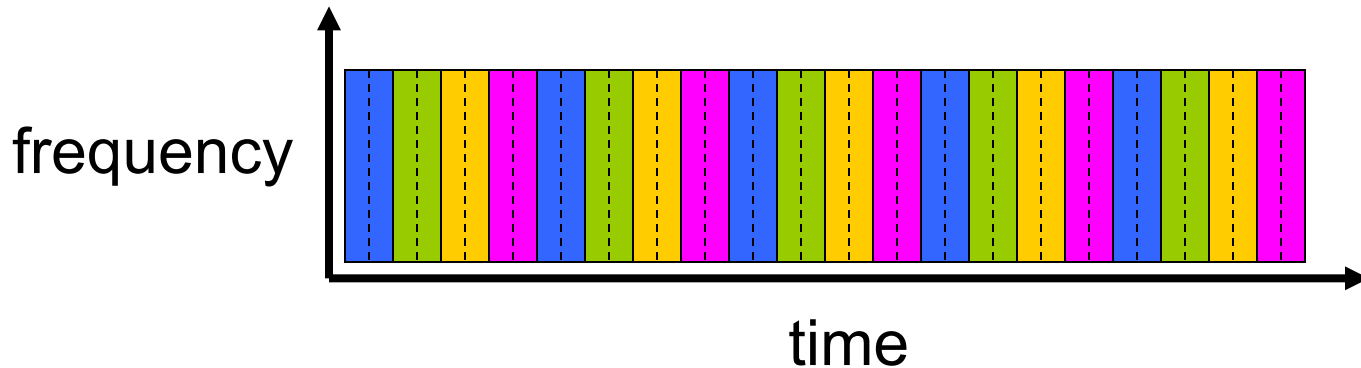
4 users



FDM



TDM



FDM Vs TDM

□ What are the tradeoffs?

- (Dis)Advantage of dividing frequency ?
- (Dis)Advantage of dividing time ?

Network Core: Packet Switching

each end-end data stream divided
into *packets*

- ❑ user A, B packets *share* network resources
- ❑ each packet uses full link bandwidth
- ❑ resources used *as needed*

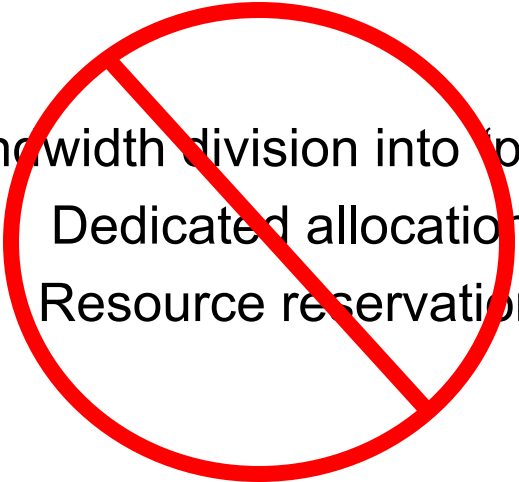
resource contention:

- ❑ aggregate resource demand can exceed amount available
 - Packets queue up
- ❑ store and forward:
packets move one hop at a time
 - Node receives complete packet before forwarding

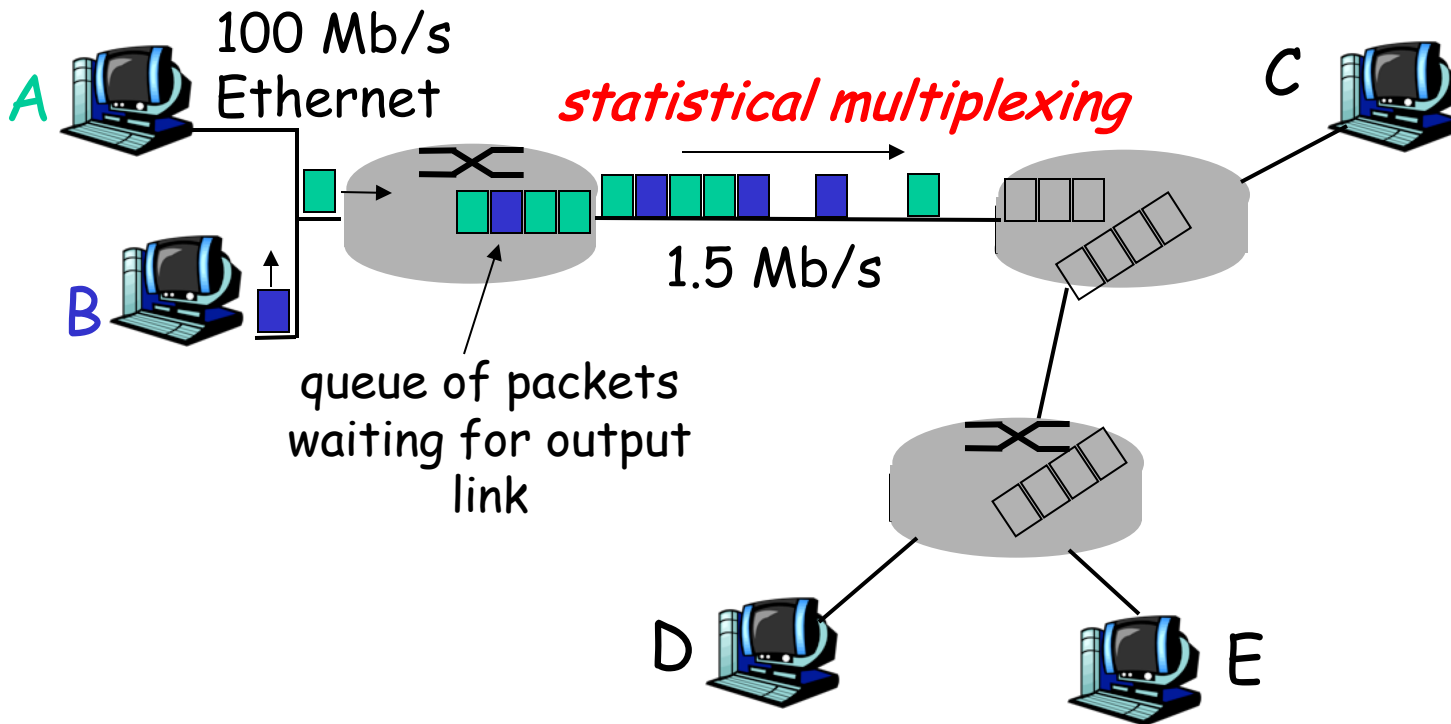
Bandwidth division into “pieces”

Dedicated allocation

Resource reservation



Packet Switching: Statistical Multiplexing



Sequence of A & B packets does not have fixed pattern, shared on demand ➡ **statistical multiplexing**.

TDM: each host gets same slot in revolving TDM frame.

Compare

Thoughts on **tradeoffs** between packet switching and circuit switching?

Which one would you take?

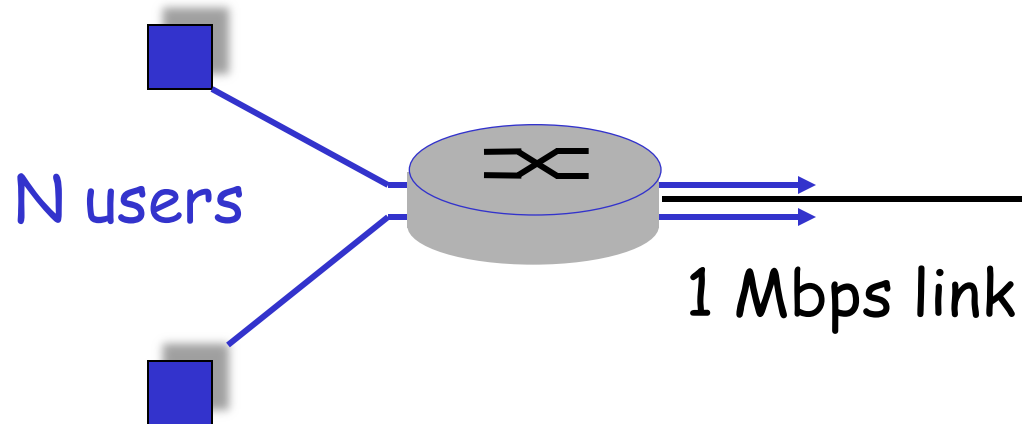
Under what circumstances?

Why?

Packet switching versus circuit switching

Packet switching allows more users to use network!

- ❑ 1 Mb/s link
- ❑ each user:
 - 100 kb/s when “active”
 - active 10% of time
- ❑ circuit-switching:
 - 10 users
- ❑ packet switching:
 - with 35 users, probability > 10 active less than .0004



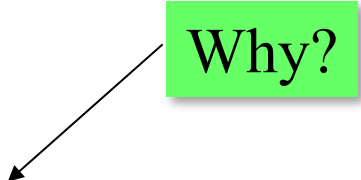
Q: how did we get value 0.0004?

Packet switching versus circuit switching

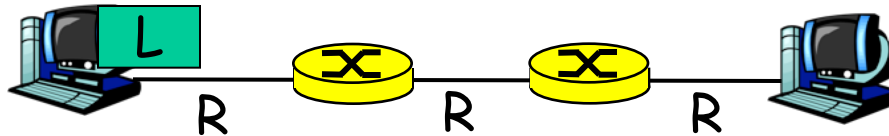
Is packet switching a “slam dunk winner?”

- ❑ Great for bursty data
 - resource sharing
 - simpler, no call setup
- ❑ Excessive congestion: packet delay and loss
 - protocols needed for reliability, congestion control
- ❑ Q: How to provide circuit-like behavior?
 - bandwidth guarantees needed for audio/video apps
 - still unsolved (chapter 7)

Why?



Packet-switching: store-and-forward



- ❑ Takes L/R seconds to transmit (push out) packet of L bits on to link of R bps
- ❑ Entire packet must arrive at router before it can be transmitted on next link: *store and forward*
- ❑ delay = $3L/R$ (assuming zero propagation delay)

Example:

- ❑ $L = 7.5$ Mbits
- ❑ $R = 1.5$ Mbps
- ❑ delay = 15 sec

} more on delay shortly ...

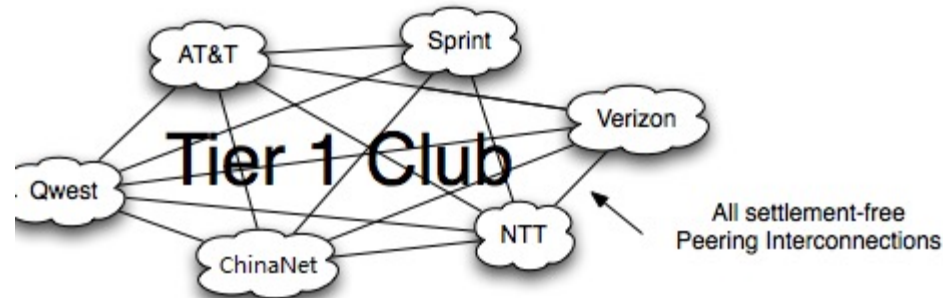
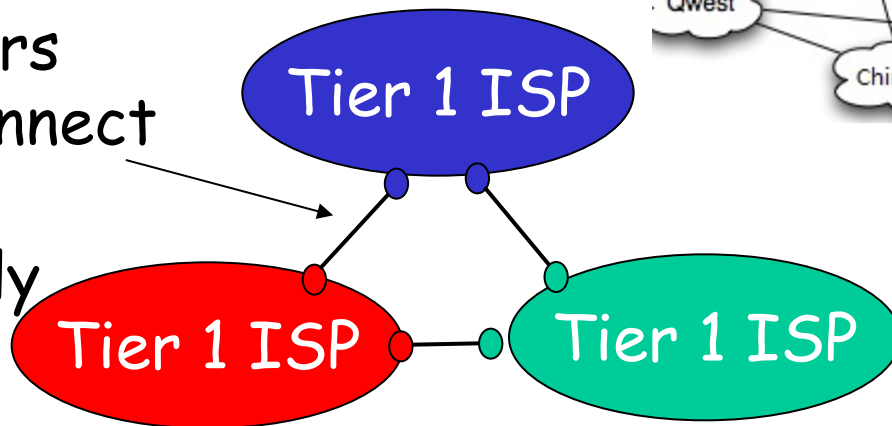
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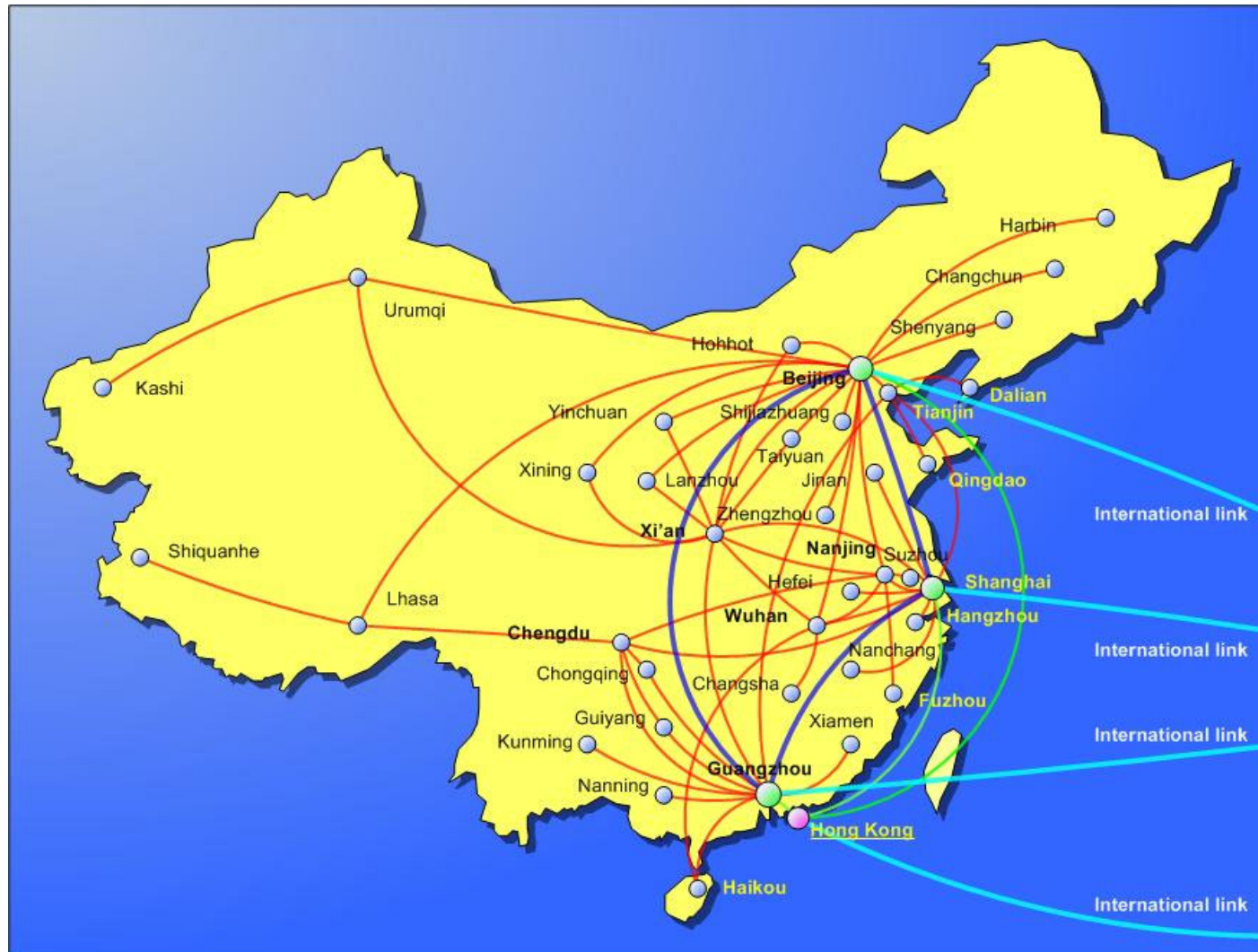
Tier-1 ISP

- ❑ can reach every other network on the Internet without purchasing IP transit or paying settlements.
- ❑ treat each other as equals
- ❑ national/international coverage (e.g., ChinaNet and ChinaGBN)

Tier-1
providers
interconnect
(peer)
privately

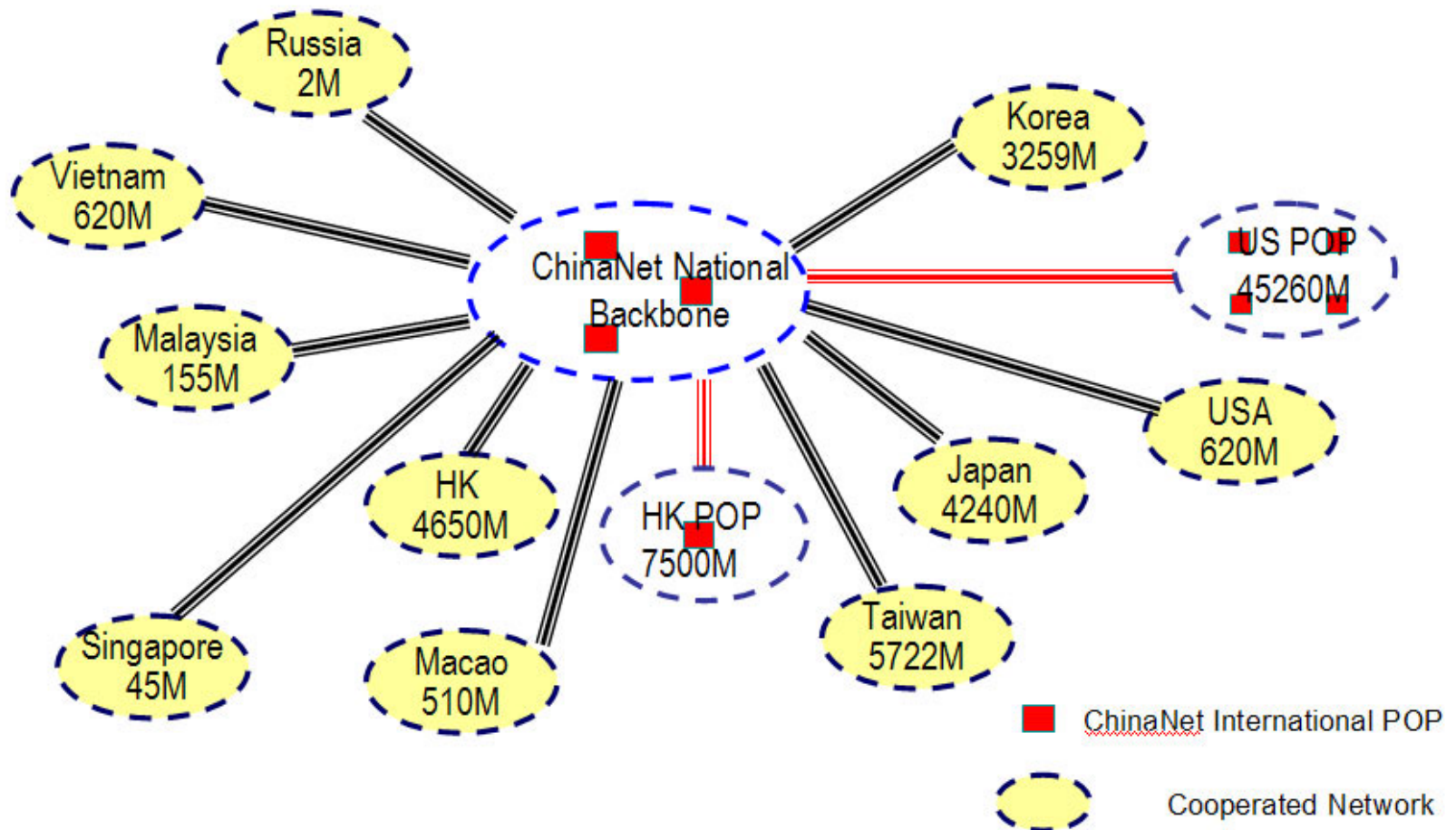


Tier-1 ISP: e.g., ChinaNet



Tier-1 ISP: e.g., ChinaNet

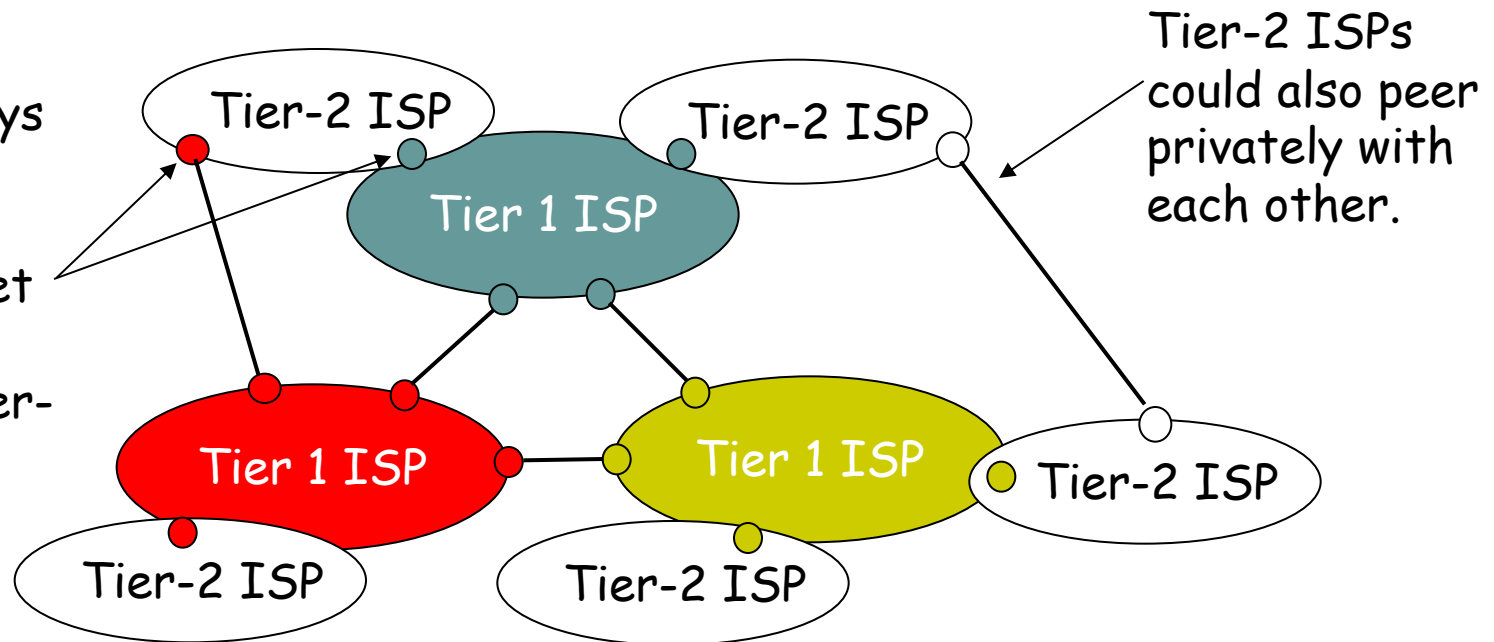
ChinaNet International Connectivity



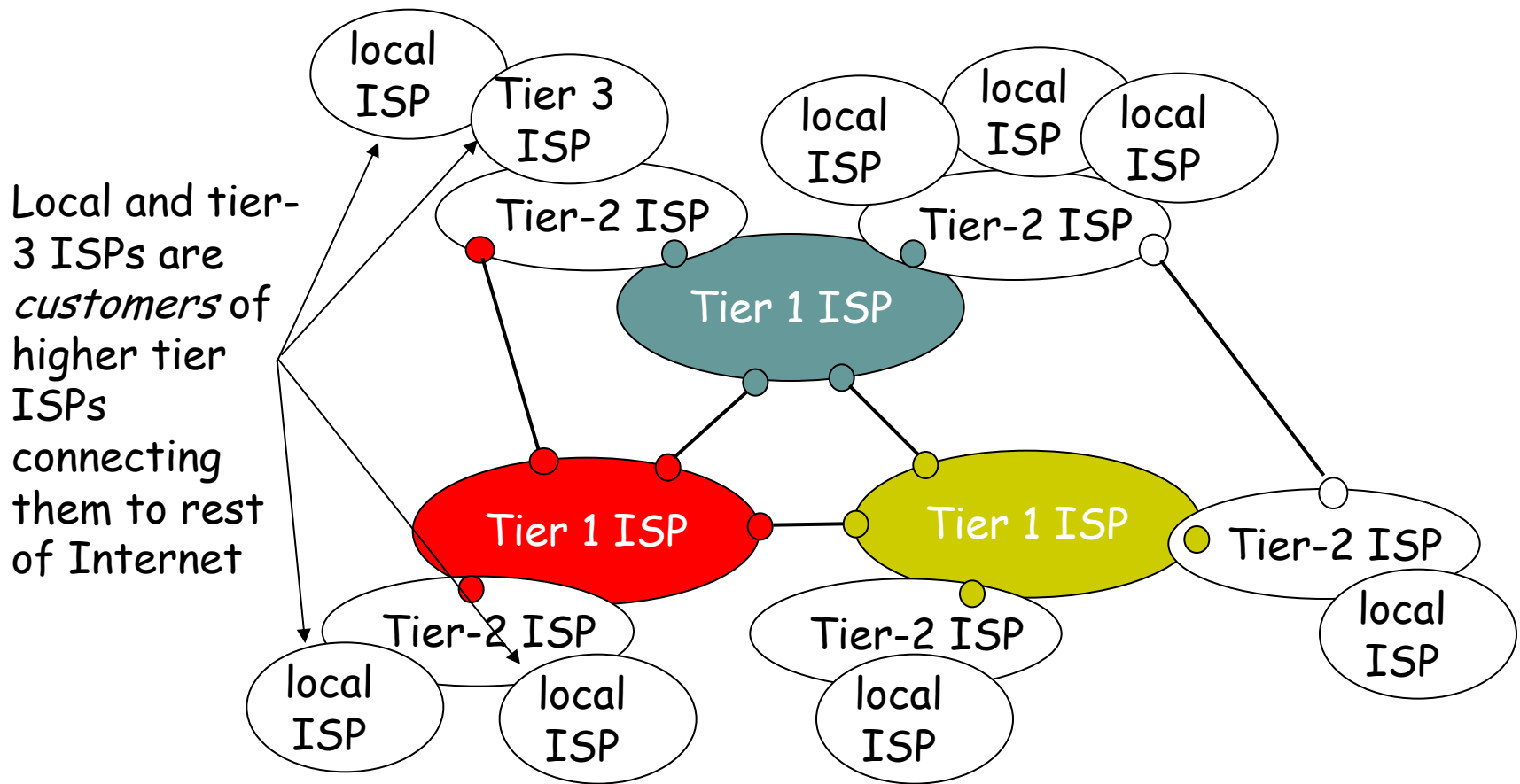
Tier-2 ISPs

- **smaller (often regional) ISPs** Connect to one or more tier-1 ISPs, possibly other tier-2 ISPs

- Tier-2 ISP pays tier-1 ISP for connectivity to rest of Internet
- tier-2 ISP is *customer* of tier-1 provider

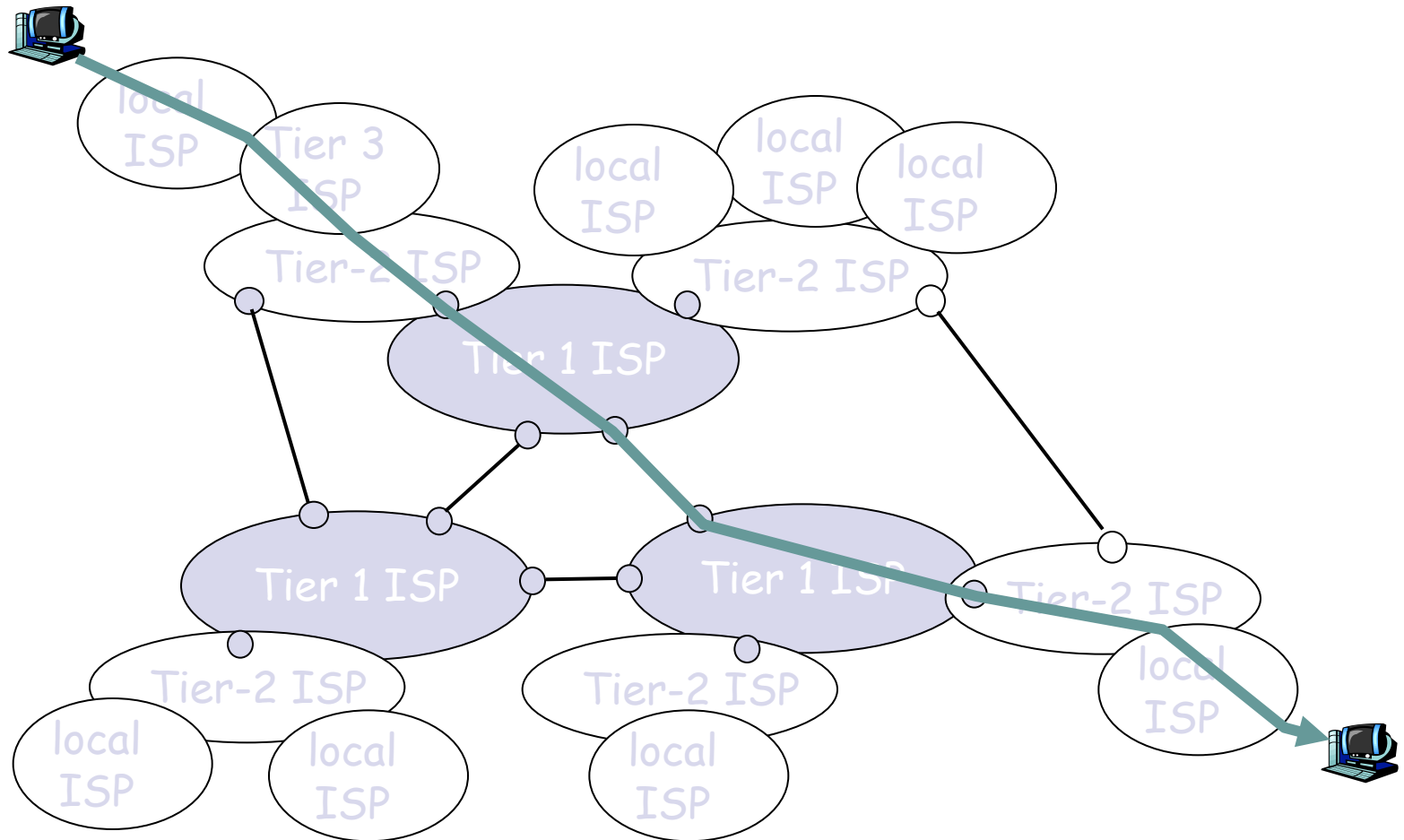


Tier-3 ISP and Local ISP



network of networks

- a packet passes through many networks!



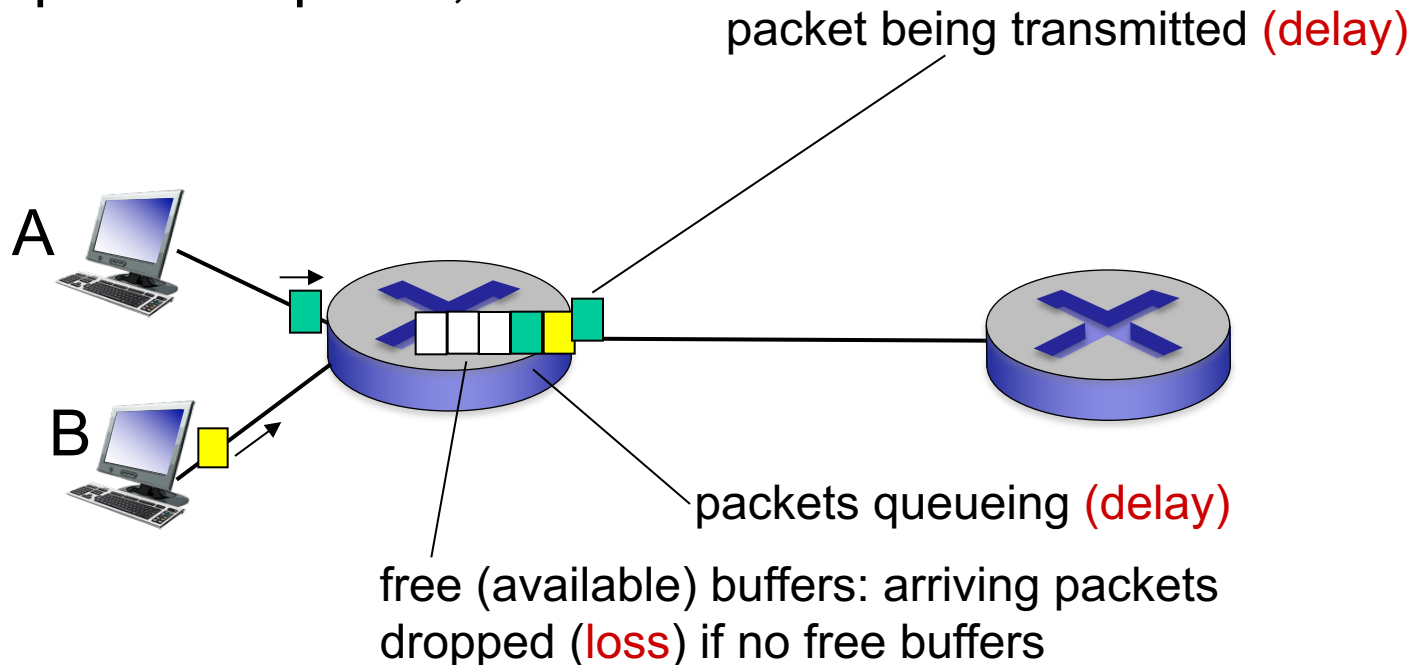
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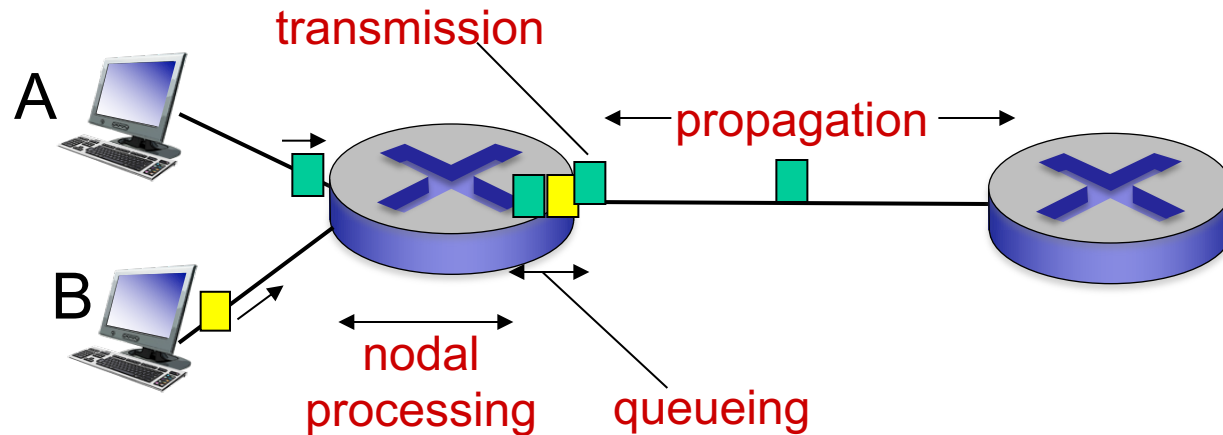
How do loss and delay occur?

packets *queue* in router buffers

- packet arrival rate to link (temporarily) exceeds output link capacity
- packets queue, wait for turn



Four sources of packet delay



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

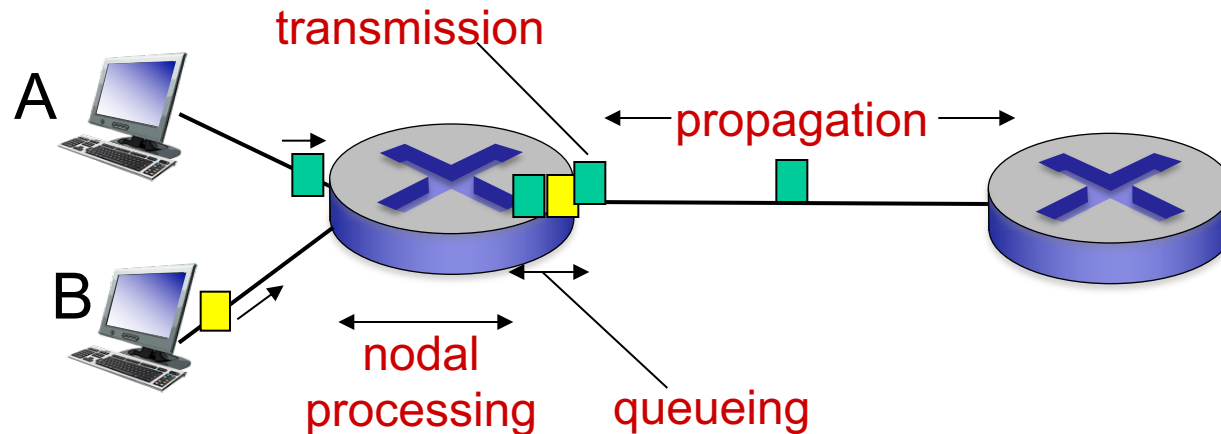
d_{proc} : nodal processing

- check bit errors
- determine output link
- typically < msec

d_{queue} : queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Four sources of packet delay



$$d_{\text{nodal}} = d_{\text{proc}} + d_{\text{queue}} + d_{\text{trans}} + d_{\text{prop}}$$

d_{trans} : transmission delay:

- L : packet length (bits)
- R : link bandwidth (bps)

■ $d_{\text{trans}} = L/R$ ← d_{trans} and d_{prop} →
very different

d_{prop} : propagation delay:

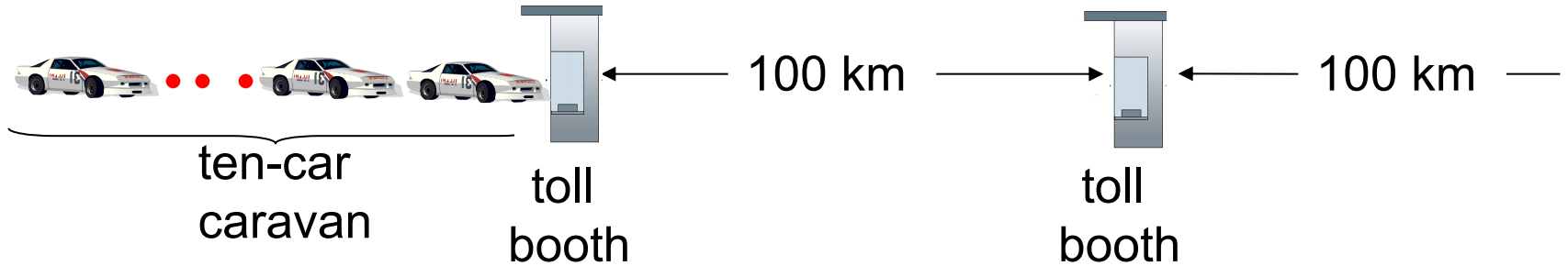
- d : length of physical link
- s : propagation speed ($\sim 2 \times 10^8$ m/sec)

■ $d_{\text{prop}} = d/s$

* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

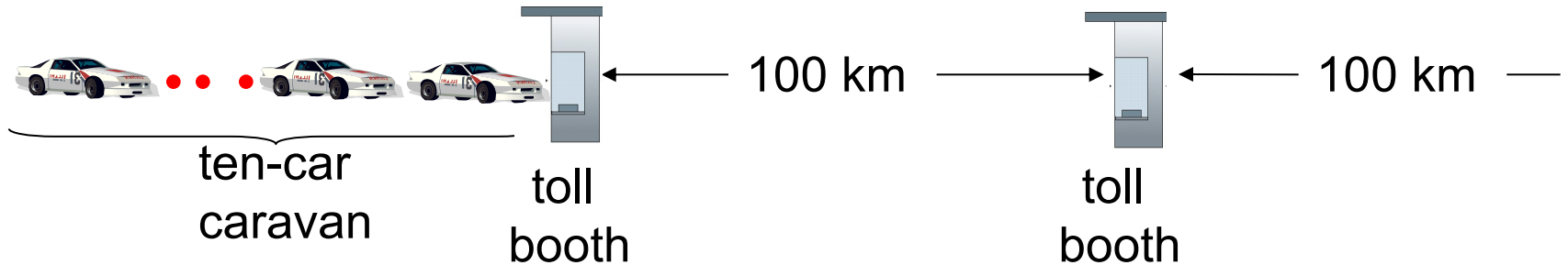
* Check out the Java applet for an interactive animation on trans vs. prop delay

Caravan analogy



- ❑ cars “propagate” at 100 km/hr
 - ❑ toll booth takes 12 sec to service car (bit transmission time)
 - ❑ car ~ bit; caravan ~ packet
 - ❑ **Q: How long until caravan is lined up before 2nd toll booth?**
- ❑ time to “push” entire caravan through toll booth onto highway = $12 \times 10 = 120$ sec
 - ❑ time for last car to propagate from 1st to 2nd toll booth:
 $100\text{km} / (100\text{km/hr}) = 1$ hr
 - ❑ **A: 62 minutes**

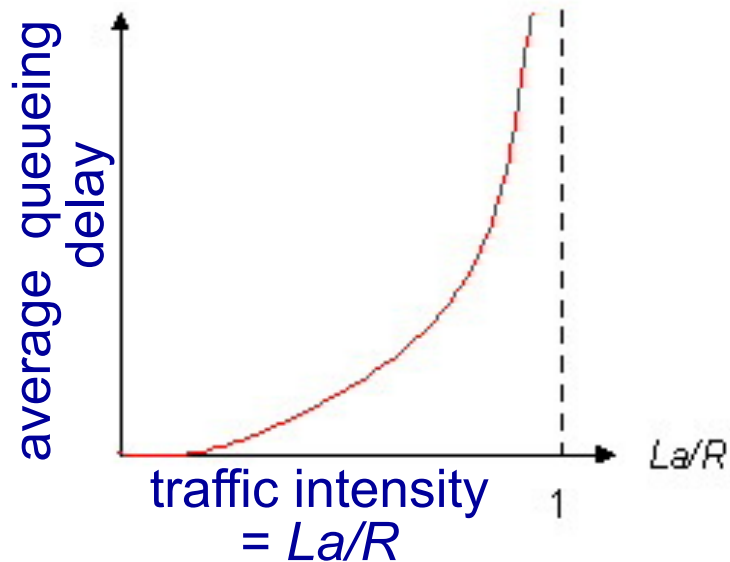
Caravan analogy (more)



- ❑ suppose cars now “propagate” at 1000 km/hr
- ❑ and suppose toll booth now takes one min to service a car
- ❑ Q: Will cars arrive to 2nd booth before all cars serviced at first booth?
 - A: Yes! after 7 min, first car arrives at second booth; three cars still at first booth

Queueing delay

- R : link bandwidth (bps)
 - L : packet length (bits)
 - a : average packet arrival rate
- rate



- $La/R \sim 0$: avg. queueing delay small
- $La/R > 1$: more
- $La/R \rightarrow 1$: avg. queueing delay large
- “work” arriving than can be serviced, average delay infinite!



$La/R \sim 0$



$La/R \rightarrow 1$

* Check online interactive animation on queueing and loss

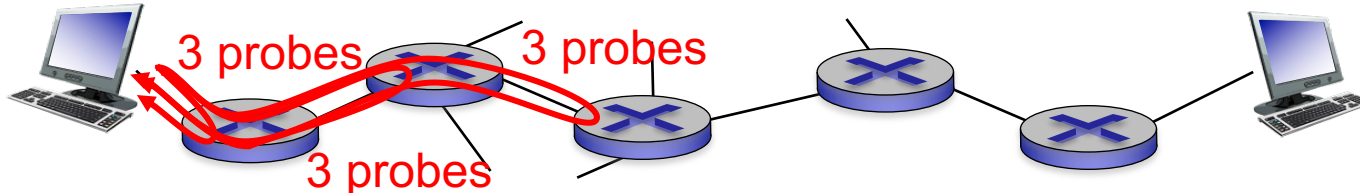
“Real” Internet delays and routes

❑ what do “real” Internet delay & loss look like?

❑ **traceroute** program: provides delay measurement from source to router along end-end Internet path towards destination.

For all i :

- sends three packets that will reach router i on path towards destination
- router i will return packets to sender
- sender times interval between transmission and reply.



“Real” Internet delays, routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr

3 delay measurements from
gaia.cs.umass.edu to cs-gw.cs.umass.edu



1 cs-gw (128.119.240.254) 1 ms 1 ms 2 ms
2 border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145) 1 ms 1 ms 2 ms
3 cht-vbns.gw.umass.edu (128.119.3.130) 6 ms 5 ms 5 ms
4 jn1-at1-0-0-19.wor.vbns.net (204.147.132.129) 16 ms 11 ms 13 ms
5 jn1-so7-0-0-0.wae.vbns.net (204.147.136.136) 21 ms 18 ms 18 ms
6 abilene-vbns.abilene.ucaid.edu (198.32.11.9) 22 ms 18 ms 22 ms
7 nycm-wash.abilene.ucaid.edu (198.32.8.46) 22 ms 22 ms 22 ms
8 62.40.103.253 (62.40.103.253) 104 ms 109 ms 106 ms
9 de2-1.de1.de.geant.net (62.40.96.129) 109 ms 102 ms 104 ms
10 de.fr1.fr.geant.net (62.40.96.50) 113 ms 121 ms 114 ms
11 renater-gw.fr1.fr.geant.net (62.40.103.54) 112 ms 114 ms 112 ms
12 nio-n2.cssi.renater.fr (193.51.206.13) 111 ms 114 ms 116 ms
13 nice.cssi.renater.fr (195.220.98.102) 123 ms 125 ms 124 ms
14 r3t2-nice.cssi.renater.fr (195.220.98.110) 126 ms 126 ms 124 ms
15 eurecom-valbonne.r3t2.ft.net (193.48.50.54) 135 ms 128 ms 133 ms
16 194.214.211.25 (194.214.211.25) 126 ms 128 ms 126 ms
17 * * *
18 * * *
19 fantasia.eurecom.fr (193.55.113.142) 132 ms 128 ms 136 ms

trans-oceanic link

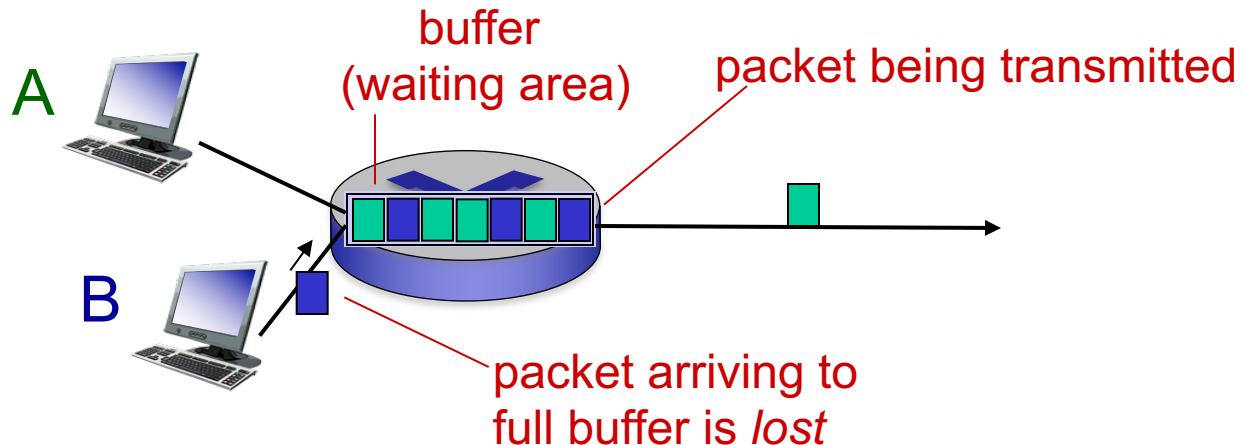


* means no response (probe lost, router not replying)

* Do some traceroutes from exotic countries at www.traceroute.org

Packet loss

- ❑ queue (aka buffer) preceding link in buffer has finite capacity
- ❑ packet arriving to full queue dropped (aka lost)
- ❑ lost packet may be retransmitted by previous node, by source end system, or not at all

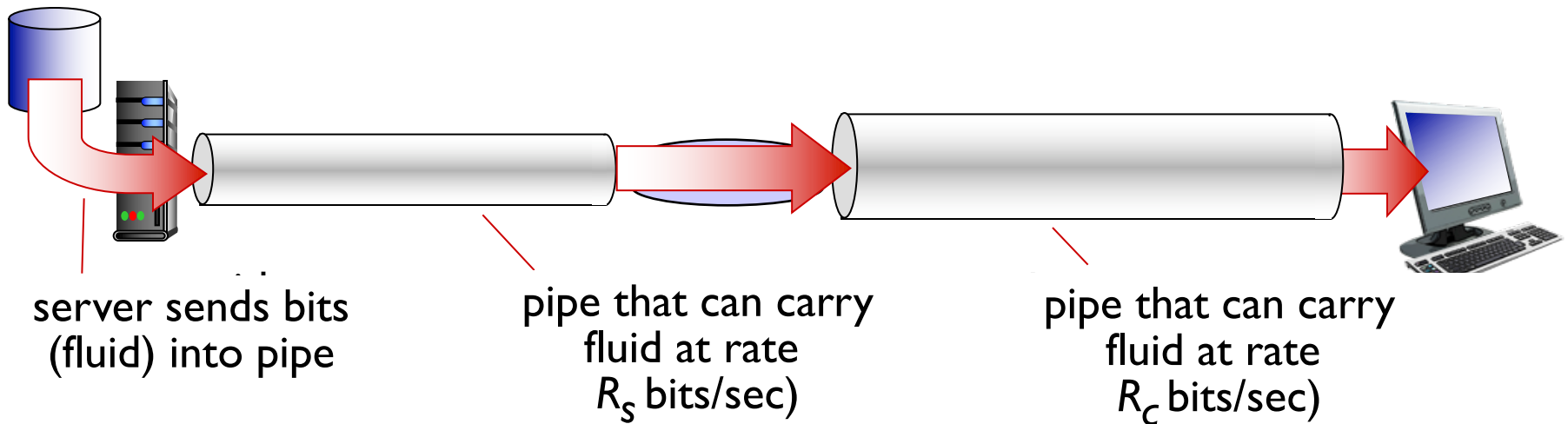


* Check out the Java applet for an interactive animation on queuing and loss

Throughput

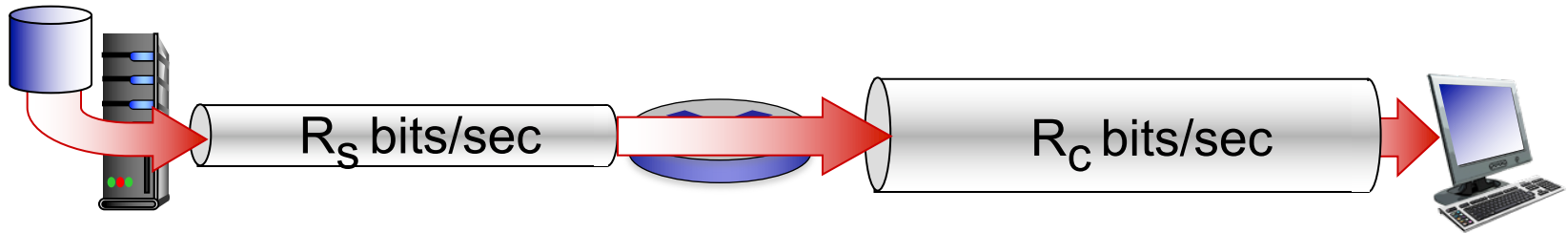
□ *throughput*: rate (bits/time unit) at which bits transferred between sender/receiver

- *instantaneous*: rate at given point in time
- *average*: rate over longer period of time

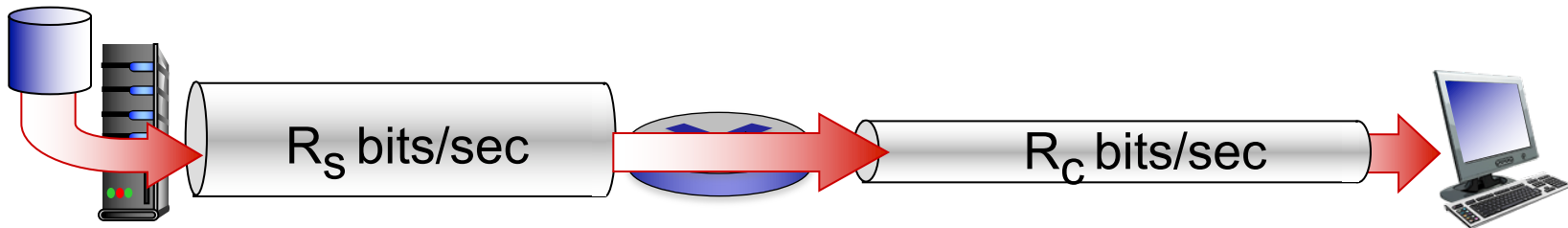


Throughput (more)

□ $R_s < R_c$ What is average end-end throughput?



■ $R_s > R_c$ What is average end-end throughput?

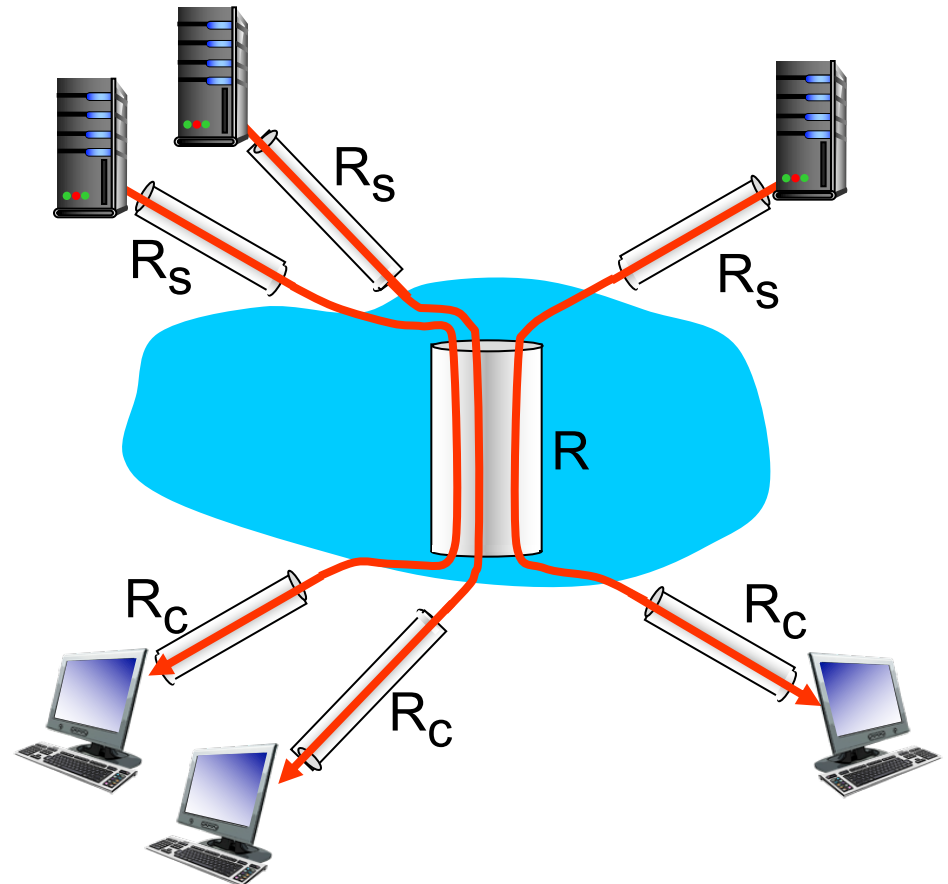


bottleneck link

link on end-end path that constrains end-end throughput

Throughput: Internet scenario

- per-connection end-end throughput:
 $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck



10 connections (fairly) share
backbone bottleneck link R bits/sec

* Check out the online interactive exercises for more
examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

Questions?

Introduction: Summary

Covered a “ton” of material!

- ❑ Internet overview
- ❑ what's a protocol?
- ❑ network edge, core, access network
 - packet-switching versus circuit-switching
- ❑ Internet/ISP structure
- ❑ performance: loss, delay
- ❑ layering and service models

You now have:

- ❑ context, overview, “feel” of networking
- ❑ more depth, detail *to follow!*

Chapter 1: roadmap

- ❑ What's Computer Network?
- ❑ Protocol layers, service models
- ❑ What's the Internet?
- ❑ Network edge
- ❑ Access net and physical media
- ❑ Network core
- ❑ Internet structure and ISPs
- ❑ Delay, loss, and throughput in packet-switched networks
- ❑ History of Internet

Chapter 1: roadmap

1.1 What *is* the Internet?

1.2 Network edge

1.3 Network core

1.4 Network access and physical media

1.5 Internet structure and ISPs

1.6 Delay & loss in packet-switched networks

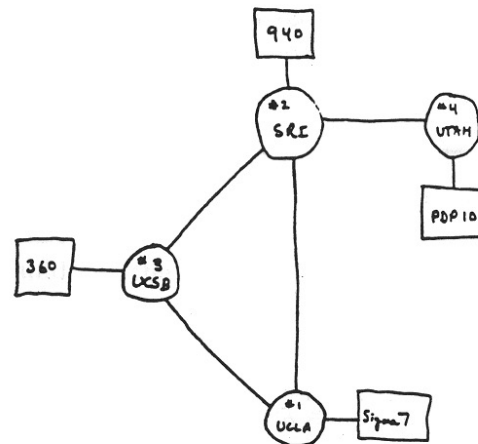
1.7 Protocol layers, service models

1.8 History

Internet History

1961-1972: Early packet-switching principles

- ❑ 1961: Kleinrock - queueing theory shows effectiveness of packet-switching
- ❑ 1964: Baran - packet-switching in military nets
- ❑ 1967: ARPAnet conceived by Advanced Research Projects Agency
- ❑ 1969: first ARPAnet node operational
- ❑ 1972:
 - ARPAnet public demonstration
 - NCP (Network Control Protocol) first host-host protocol
 - first e-mail program
 - ARPAnet has 15 nodes



THE ARPA NETWORK

Internet History

1972-1980: Internetworking, new and proprietary nets

- ❑ 1970: ALOHAnet satellite network in Hawaii
- ❑ 1974: Cerf and Kahn - architecture for interconnecting networks
- ❑ 1976: Ethernet at Xerox PARC
- ❑ late70's: proprietary architectures: DECnet, SNA, XNA
- ❑ late 70's: switching fixed length packets (ATM precursor)
- ❑ 1979: ARPAnet has 200 nodes

Cerf and Kahn's internetworking principles:

- minimalism, autonomy - no internal changes required to interconnect networks
- best effort service model
- stateless routers
- decentralized control

define today's Internet architecture

Internet History

1980-1990: new protocols, a proliferation of networks

- ❑ 1983: deployment of TCP/IP
- ❑ 1982: smtp e-mail protocol defined
- ❑ 1983: DNS defined for name-to-IP-address translation
- ❑ 1985: ftp protocol defined
- ❑ 1988: TCP congestion control
- ❑ new national networks: Csnet, BITnet, NSFnet, Minitel
- ❑ 100,000 hosts connected to confederation of networks

Internet History

1990, 2000's: commercialization, the Web, new apps

- ❑ Early 1990's: ARPAnet decommissioned
- ❑ 1991: NSF lifts restrictions on commercial use of NSFnet (decommissioned, 1995)
- ❑ early 1990s: Web
 - hypertext [Bush 1945, Nelson 1960's]
 - HTML, HTTP: Berners-Lee
 - 1994: Mosaic, later Netscape
 - late 1990's: commercialization of the Web

Late 1990's – 2000's:

- ❑ more killer apps: instant messaging, P2P file sharing
- ❑ network security to forefront
- ❑ est. 50 million host, 100 million+ users
- ❑ backbone links running at Gbps